

PreOne Master PRD

Enterprise Preschool Operating System

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1. Introduction

1.1 Purpose

या दस्तऐवजाचा उद्देश PreOne Platform मधील सर्व Functional आणि Business Requirements स्पष्टपणे define करणे आहे. हा Master PRD product development lifecycle मधील authoritative reference document आहे, जो engineering team ला implementation scope देतो, QA team ला acceptance criteria देतो, आणि business stakeholders ला commitment lock देतो.

हा document Vision v1.0, BPM v1.0 आणि BRC v1.0 चा natural successor आहे. Vision ने "कुठे जायचे" सांगितले, BPM ने "कसे चालवायचे" detailed केले, BRC ने "कोणते rules enforce करायचे" list केले, आणि Master PRD आता "काय build करायचे" exactly define करते. प्रत्येक requirement traceable आहे — BR वरून FR पर्यंत, FR वरून BRC rule पर्यंत, आणि शेवटी acceptance criterion पर्यंत.

1.2 Scope

PreOne खालील 13 Business Domains कव्हर करेल. हे domains DDD philosophy नुसार bounded contexts म्हणून design केले आहेत — प्रत्येक domain ची स्वतःची data model, business rules, workflows आणि APIs आहेत:

#	Domain	Scope Summary	Primary Persona
1	Marketing & CRM	Lead capture, counselling, follow-ups, conversion analytics	Reception, Owner
2	Admissions	Online applications, document verification, workflow automation	Reception, Principal
3	Student Lifecycle	Allocation, transfers, archives, graduation, alumni	Principal, Coordinator
4	Academics	Curriculum, milestones, observations, portfolios, report cards	Teacher, Coordinator

#	Domain	Scope Summary	Primary Persona
5	Daily Operations	Attendance, daily sheet, meals, nap, activities, incidents	Teacher
6	Parent Communication	Announcements, events, chat, SMS, WhatsApp, email	Parent, Teacher
7	Finance	Fee plans, invoices, payments, receipts, expenses, refunds	Accounts, Owner
8	Inventory & Procurement	Purchase requests, POs, GRN, issues, returns, audits	Accounts, Coordinator
9	Human Resources	Staff, recruitment, leave, payroll, training, performance	Principal, Owner
10	Administration	Assets, maintenance, visitors, compliance, meal plans	Principal, Coordinator
11	Reports & Analytics	Real-time dashboards, KPIs, trends, predictions	Owner, Principal
12	Settings	Configurations, templates, notification rules, integrations	Principal, Owner
13	Platform Management	Multi-tenant SaaS layer, billing, subscriptions, audit	Platform Admin

1.3 Document Conventions

हा document काही standard conventions follow करतो जे वाचणं आणि reference करणं सोपं करतात. Requirement IDs (BR-001, FR-001, AC-001) stable identifiers आहेत जे future versions मध्ये change होणार नाहीत — जर कोणती requirement deprecate होईल तर ID retired केली जाईल, reuse

होणार नाही. Priority levels: P0 (must-have for v1.0 GA), P1 (should-have for v1.0 GA), P2 (nice-to-have, v1.1+ candidate). Status: Draft, Reviewed, Approved, Frozen.

Marathi + Hindi + English mixed language convention हा document मध्ये जाणीवपूर्वक वापरला आहे — business context Marathi/Hindi मध्ये explain केला आहे ज्यामुळे Indian preschool stakeholders ना native resonance मिळतो, आणि technical terms English मध्ये ठेवले आहे ज्यामुळे engineering team ची implementation clarity राहते. Code snippets, field names, API paths, error codes हे always English मध्ये आहेत.

1.4 Related Documents

Document	Version	Status	Role
PreOne Vision Document	1.0	Frozen	Strategic direction, target market, success metrics
PreOne BPM (Business Process Map)	1.0	Frozen	Cross-domain process flows, 13 domain workflows
PreOne BRC (Business Rules Catalog)	1.0	Frozen	176 rules across 12 categories, 14-field schema each
PreOne ADR (Architecture Decision Records)	1-12	Frozen	Tech stack, multi-tenant model, identity, AI architecture
PreOne Database Schema	v2	Frozen	69 models across 13 modules, Prisma schema reference
PreOne Master PRD (this document)	1.0	Product Freeze	Functional + business requirements, acceptance criteria

2. Product Overview

2.1 Product Vision

PreOne हे AI-ready Enterprise Preschool Operating System आहे जे Preschool मधील सर्व Business Processes एका एकात्मिक Platform वर आणते. ही vision statement तीन strategic choices represent करते: "AI-ready" म्हणजे AI हा bolted-on feature नाही तर platform च्या architecture मध्ये embedded आहे; "Enterprise" म्हणजे multi-school chains साठी scalable, RBAC, audit, compliance built-in; "Operating System" म्हणजे केवळ ERP नाही तर daily users चा primary workspace.

हे vision PreOne ला "yet another school ERP" पासून स्पष्टपणे वेगळे करते. Generic school ERPs फक्त records ट्रॅक करतात (attendance, marks, fees) पण preschool-specific workflows (daily child timeline, learning observations, activity-based curriculum, age-appropriate milestones) त्यांच्या data model मध्ये fundamentally fit होऊ शकत नाहीत. PreOne हे ground-up preschool-specific design केलेले आहे.

2.2 Objectives

PreOne चे पाच primary objectives खालीलप्रमाणे आहेत. हे एकमेकांशी aligned आहेत — कोणताही single objective स्वतःचा priority claim करू शकत नाही, त्यांचा combined effect PreOne चे product character define करतो:

- **Paperless Operations:** Preschool मधील प्रत्येक record — admission form, attendance register, fee receipt, observation note, stock book — digital format मध्ये capture होईल. Paper usage 95%+ कमी होईल.
- **Mobile-first Experience:** Teacher app (tablet-optimized) आणि Parent app (phone-first) हे primary surfaces आहेत. Admin portal desktop-first पण responsive — owners travel करताना dashboard check करू शकतील.
- **Enterprise-grade Security:** JWT auth, MFA, RBAC, audit trail, AES-256 encryption at-rest, TLS 1.3 in-transit, tenant isolation, DPDP Act 2023 compliance, ISO 27001 alignment.
- **Multi-School SaaS:** True multi-tenant architecture — single codebase, school_id isolation at every table, per-school configurations, cross-school analytics for chain owners. 10,000+ schools target.
- **AI-ready Architecture:** AI capabilities embedded across platform — observation assistant, daily sheet auto-draft, report card AI paragraph generation, fee default prediction, parent reply composer, growth milestone prediction.

2.3 Target Users

PreOne चे 7 primary personas आहेत ज्यांच्यासाठी product design केले गेले आहे. प्रत्येक persona ची distinct context, goals, pain points आणि success metrics आहेत. Feature prioritization चा primary framework हे persona coverage matrix आहे — जे feature 2+ personas च्या needs address करते ते P0.

Persona	Role Context	Primary Device	Success Metric
Preschool Owner	Business owner / investor	Desktop + Mobile	Revenue visibility + branch ROI
Principal	Academic + operational head	Desktop + Mobile	Operations smooth + parent satisfaction
Coordinator	Multi-class supervisor	Tablet + Desktop	Teacher productivity + class outcomes
Teacher	Classroom execution	Tablet (primary)	Less admin time + better observations
Parent	Child's guardian	Mobile (primary)	Real-time child visibility + easy fee pay
Accounts Team	Finance operations	Desktop	Fee collection % + reconciliation accuracy
Reception	Front desk	Desktop + Tablet	Lead response time + admission conversion

2.4 Product Pillars

PreOne चे design decisions पाच pillars शी aligned असावे लागतात. हे pillars design review चा framework आहेत आणि जेव्हा feature tradeoffs येतात तेव्हा tie-breaker म्हणून वापरले जातात:

- **Preschool First:** Architecture decisions पासून UI copy पर्यंत preschool context reflect होईल — activities वर subjects, milestones वर grades, observations वर exams.
- **Mobile First:** Teacher tablet app आणि Parent phone app हे primary surfaces. Desktop portal secondary.

- **Enterprise Ready:** Multi-school, multi-branch, RBAC, audit, security, SaaS architecture built-in from day 1.
- **AI Ready:** AI capabilities platform च्या प्रत्येक layer मध्ये embedded — data model, workflows, notifications, reports, analytics.
- **Simple Yet Powerful:** Progressive disclosure — small preschool सहज वापरू शकेल, enterprise chains सहज scale करू शकतील.

2.5 Architecture Snapshot

PreOne चे architecture Domain-Driven Design (DDD) philosophy वर आधारित आहे. 13 business domains हे 13 bounded contexts आहेत ज्यांची स्वतःची data models, business rules, workflows आणि APIs आहेत. Cross-domain communication event-driven architecture द्वारे होते — Identity, Notification, Audit, Approval, Reporting हे 5 cross-domain flows सर्व domains ला tie करतात.

Tech stack: Next.js 16 (frontend + API routes), Prisma ORM (PostgreSQL multi-tenant), Redis (cache + queue), S3-compatible object storage (file uploads), Razorpay/Cashfree (payments), MSG91/Twilio (SMS), WhatsApp Cloud API, SES/SendGrid (email), LLM APIs (AI features). Multi-tenant isolation strategy: shared database with school_id discriminator on every table + row-level security policies.

3. Business Requirements (BR)

Business Requirements हे "What the business needs" याचे उत्तर देतात. हे requirements technical implementation वर depend करत नाहीत — ते business outcomes आहेत जे PreOne ने achieve करायचे आहेत. प्रत्येक BR चा unique ID, clear objective, measurable business value, priority, owner, related domains, dependencies आणि success metrics आहेत.

खालील 30 Business Requirements 13 domains वर distributed आहेत. Priority distribution: P0 (must-have v1.0) = 18, P1 (should-have v1.0) = 9, P2 (nice-to-have v1.1+) = 3. हे distribution development focus देतो — v1.0 चा scope फक्त P0 + P1 आहे, P2 explicitly v1.1+ मध्ये push केले आहे.

BR ID	Title	Priority	Owner Domain	Dependencies	Success Metrics
BR-001	Student Admission	P0	Admissions	Marketing, Finance	Lead-to-enrolled conversion $\geq 25\%$, admission TAT ≤ 7 days
BR-002	Parent Communication	P0	Communication	Daily Ops, Academics	Parent DAU/MAU $\geq 60\%$, message response time ≤ 4 hours
BR-003	Daily Operations Digital	P0	Daily Ops	Academics	Daily sheet completion $\geq 95\%$, paper usage reduction 90%+
BR-004	Fee Management	P0	Finance	Admissions	Fee collection $\geq 92\%$, online payment adoption $\geq 70\%$
BR-005	Inventory Management	P0	Inventory	Finance, Admin	Stock accuracy $\geq 98\%$, wastage reduction 30%+
BR-006	Multi-School SaaS	P0	Platform	Settings	10,000+ schools scale, 99.9% uptime, NRR $\geq 110\%$

BR ID	Title	Priority	Owner Domain	Dependencies	Success Metrics
BR-007	Lead Capture & CRM	P0	Marketing	Admissions	Lead response ≤ 30 min, lead-to-visit $\geq 40\%$
BR-008	Student Lifecycle	P0	Student Lifecycle	Admissions, Academics	Zero data loss on transfer, alumni access retained 5 years
BR-009	Academic Observations	P0	Academics	Daily Ops	Observation frequency ≥ 3 /child/week, parent visibility 100%
BR-010	HR & Payroll	P0	HR	Finance	Payroll accuracy 100%, leave transparency 100%
BR-011	Compliance Management	P0	Admin	Settings, Platform	100% compliance adherence, 0 audit findings
BR-012	Real-time Dashboards	P0	Reports	All domains	Dashboard DAU $\geq 80\%$ management, decision speed 5x faster

BR ID	Title	Priority	Owner Domain	Dependencies	Success Metrics
BR-013	Notification Engine	P0	Platform	Communication	Notification delivery $\geq 99\%$, opt-out rate $\leq 2\%$
BR-014	Audit Trail	P0	Platform	All domains	100% actions logged, log retention 7 years
BR-015	Mobile Apps	P0	Platform	All domains	App store rating ≥ 4.5 , crash-free $\geq 99.5\%$
BR-016	Multi-language Support	P1	Platform	Settings	3 languages v1.0 (English, Marathi, Hindi), 8 by year 3
BR-017	Visitor Management	P1	Admin	Communication	100% visitor check-in digital, child pickup auth 100%
BR-018	Asset Management	P1	Admin	Finance	Asset tracking 100%, maintenance schedule adherence $\geq 90\%$
BR-019	Event Management	P1	Communication	Academics, Finance	Event RSVP rate $\geq 75\%$, event revenue tracking 100%

BR ID	Title	Priority	Owner Domain	Dependencies	Success Metrics
BR-020	Report Cards	P1	Academics	Reports	Report card generation ≤ 2 hours/class, parent ack $\geq 90\%$
BR-021	Refund Processing	P1	Finance	Admissions	Refund TAT ≤ 7 days, 100% audit trail
BR-022	Bulk Import/Export	P1	Platform	All domains	10,000 records ≤ 5 min, 0 data corruption
BR-023	Custom Workflows	P1	Settings	All domains	Workflow builder adoption $\geq 30\%$ schools
BR-024	Biometric Attendance	P1	Daily Ops	HR	Biometric adoption $\geq 40\%$ staff, spoof rate 0%
BR-025	AI Observation Assistant	P1	Academics	Daily Ops	AI suggestion acceptance $\geq 40\%$, teacher time saved $\geq 30\%$
BR-026	Parent Feedback	P1	Communication	Reports	NPS ≥ 50 , feedback response rate $\geq 30\%$
BR-027	Third-party Integrations	P2	Platform	Settings	10+ integrations v1.1+

BR ID	Title	Priority	Owner Domain	Dependencies	Success Metrics
BR-028	Transport Management	P2	Admin	Communication	Live tracking, route optimization
BR-029	Marketplace	P2	Platform	Finance	10+ vendors, 5% conversion
BR-030	Voice-based Daily Sheet	P2	Daily Ops	Academics	Voice entry adoption ≥20% teachers

3.1 Business Requirement Template

प्रत्येक Business Requirement खालील template नुसार structured आहे. हा template future BR additions साठी reference format आहे:

Field	Description	Example
Business Requirement ID	Stable identifier, format BR-XXX	BR-001
Title	Concise business outcome name	Student Admission
Objective	What the system shall do (1-2 sentences)	System shall support complete digital admission lifecycle.
Business Value	Why this matters to the business	Faster admissions, reduced paperwork, better conversion tracking.
Priority	P0/P1/P2 with rationale	P0 — core revenue-driving workflow
Owner	Domain accountable for delivery	Admissions Domain
Related Domains	Other domains impacted	Marketing, Finance

Field	Description	Example
Dependencies	Predecessor requirements or systems	Marketing lead data, Finance invoice engine
Success Metrics	Measurable acceptance criteria	Lead-to-enrolled $\geq 25\%$, TAT ≤ 7 days

4. Functional Requirements (FR)

Functional Requirements हे "System काय करेल?" याचे उत्तर देतात. हे requirements business requirements ला actionable system behavior मध्ये translate करतात. प्रत्येक FR चा unique ID, description, actors, preconditions, workflow, inputs, outputs, validations, exceptions, related business rules आणि priority आहे.

खालील 60 Functional Requirements 13 domains वर distributed आहेत. प्रत्येक FR detailed implementation spec चा anchor point आहे — engineering team त्याच्या आधारे user stories आणि technical tasks derive करेल. Priority distribution: P0 = 36, P1 = 18, P2 = 6.

FR ID	Title	Domain	Priority	Related BR
FR-001	Lead Capture	Marketing	P0	BR-007
FR-002	Lead Follow-up Tracking	Marketing	P0	BR-007
FR-003	Counselling Session Booking	Marketing	P0	BR-007
FR-004	Lead Conversion Analytics	Marketing	P1	BR-007
FR-005	Campaign Tracking	Marketing	P2	BR-007
FR-006	Online Application Form	Admissions	P0	BR-001
FR-007	Document Verification	Admissions	P0	BR-001
FR-008	Admission Workflow	Admissions	P0	BR-001

FR ID	Title	Domain	Priority	Related BR
FR-009	Offer Letter Generation	Admissions	P0	BR-001
FR-010	Student Record Creation	Admissions	P0	BR-001
FR-011	Class & Section Allocation	Student Lifecycle	P0	BR-008
FR-012	Student Transfer	Student Lifecycle	P0	BR-008
FR-013	Student Promotion	Student Lifecycle	P0	BR-008
FR-014	Student Archives & Alumni	Student Lifecycle	P1	BR-008
FR-015	Curriculum Management	Academics	P0	BR-009
FR-016	Observation Recording	Academics	P0	BR-009
FR-017	Milestone Tracker	Academics	P0	BR-009
FR-018	Portfolio Generation	Academics	P1	BR-009
FR-019	Report Card Generation	Academics	P1	BR-020
FR-020	Attendance Marking	Daily Ops	P0	BR-003
FR-021	Daily Sheet Creation	Daily Ops	P0	BR-003
FR-022	Meal Tracking	Daily Ops	P0	BR-003
FR-023	Activity Logging	Daily Ops	P0	BR-003
FR-024	Incident Reporting	Daily Ops	P0	BR-003

FR ID	Title	Domain	Priority	Related BR
FR-025	Nap & Bathroom Tracking	Daily Ops	P0	BR-003
FR-026	Parent App Timeline	Communication	P0	BR-002
FR-027	Two-way Chat	Communication	P0	BR-002
FR-028	Announcements & Broadcasts	Communication	P0	BR-002
FR-029	Event Calendar & RSVP	Communication	P1	BR-019
FR-030	Fee Plan Configuration	Finance	P0	BR-004
FR-031	Invoice Generation	Finance	P0	BR-004
FR-032	Payment Processing	Finance	P0	BR-004
FR-033	Receipt Generation	Finance	P0	BR-004
FR-034	Refund Processing	Finance	P1	BR-021
FR-035	Expense Tracking	Finance	P1	BR-004
FR-036	Purchase Request	Inventory	P0	BR-005
FR-037	Purchase Order	Inventory	P0	BR-005
FR-038	Goods Receipt Note (GRN)	Inventory	P0	BR-005
FR-039	Stock Issue & Returns	Inventory	P0	BR-005
FR-040	Staff Recruitment	HR	P1	BR-010
FR-041	Leave Management	HR	P0	BR-010

FR ID	Title	Domain	Priority	Related BR
FR-042	Payroll Processing	HR	P0	BR-010
FR-043	Performance Review	HR	P1	BR-010
FR-044	Asset Management	Admin	P1	BR-018
FR-045	Visitor Management	Admin	P1	BR-017
FR-046	Maintenance Scheduling	Admin	P1	BR-018
FR-047	Compliance Dashboard	Admin	P0	BR-011
FR-048	Operational Dashboards	Reports	P0	BR-012
FR-049	Custom Report Builder	Reports	P1	BR-012
FR-050	Predictive Analytics	Reports	P2	BR-012
FR-051	School Configuration	Settings	P0	BR-006
FR-052	Notification Rules	Settings	P0	BR-013
FR-053	Workflow Builder	Settings	P1	BR-023
FR-054	Tenant Onboarding	Platform	P0	BR-006
FR-055	Subscription & Billing	Platform	P0	BR-006
FR-056	Audit Log & Compliance	Platform	P0	BR-014
FR-057	User Authentication	Platform	P0	BR-015

FR ID	Title	Domain	Priority	Related BR
FR-058	Role-Based Access Control	Platform	P0	BR-015
FR-059	Multi-language UI	Platform	P1	BR-016
FR-060	Bulk Import/Export	Platform	P1	BR-022

4.1 Sample FR Detail — FR-001 User Authentication

खालील FR-001 detailed template दर्शवते कसा प्रत्येक FR structured आहे. हेच template सर्व 60 FRs साठी apply होतो, पण space constraints मुळे या document मध्ये फक्त summary table + एक sample detail दिले आहेत. Implementation team ला PRD companion wiki मध्ये प्रत्येक FR चे full detail available असेल.

Requirement ID	FR-001
Title	User Authentication
Description	User shall login using Mobile Number + OTP / Password. MFA support for admin roles.
Actors	Parent, Teacher, Admin
Preconditions	Active account exists; mobile number verified
Workflow	Open App → Login → OTP Verification → (MFA for admin) → Authentication → Dashboard
Inputs	Mobile number (mandatory), OTP (6 digits, 5-min expiry) OR Password, MFA token (admin only)
Outputs	JWT access token, refresh token, user profile, role permissions, dashboard
Validations	Mobile mandatory + 10 digits; OTP expiry 5 min; max 3 retries; MFA required for admin roles
Exceptions	Lockout after 5 failed attempts → 30-min cooldown + support contact; OTP delivery fail → fallback channel (SMS if push fails)

Business Rules	BR-015, BR-011; BRC R-AUTH-001 (lockout policy), R-AUTH-002 (MFA mandate), R-SEC-001 (token expiry)
Priority	P0

4.2 Functional Requirement Template

प्रत्येक Functional Requirement खालील template नुसार structured आहे. Engineering team या template चा वापर करून user stories, technical tasks आणि test cases derive करतो:

Field	Description	Mandatory
Requirement ID	Stable identifier FR-XXX	Yes
Title	Concise feature name	Yes
Description	What the system shall do (1-3 sentences)	Yes
Actors	User roles involved	Yes
Preconditions	What must be true before FR executes	Yes
Workflow	Step-by-step flow (text or visual)	Yes
Inputs	Input fields with types	Yes
Outputs	System outputs (records, notifications, files)	Yes
Validations	Field-level + business validations	Yes
Exceptions	Error cases and handling	Yes
Business Rules	Related BR IDs and BRC rule IDs	Yes
Priority	P0/P1/P2 with rationale	Yes

5. Non-Functional Requirements (NFR)

Non-Functional Requirements हे System Quality Attributes define करतात. हे requirements "how well" the system should perform address करतात — performance, availability, security, scalability. हे requirements architecture आणि infrastructure decisions ला drive करतात.

हे ADR मध्ये freeze केलेल्या Identity आणि Authorization मॉडेलशी सुसंगत असेल. खालील 8 NFR categories each with measurable targets — performance, availability, scalability, security, reliability, maintainability, usability, localization. प्रत्येक NFR चा measurement method आणि acceptance method देखील defined आहे ज्यामुळे QA team ते verify करू शकतो.

5.1 Performance

Requirement	Target	Measurement	Acceptance Method
Login response time	≤2 seconds (95th percentile)	APM monitoring + synthetic tests	Load test 1000 concurrent users; 95% ≤2s
Dashboard load time	≤3 seconds (95th percentile)	RUM + synthetic	Real user monitoring 30 days; 95% ≤3s
Search response time	≤1 second (95th percentile)	APM + query log	Search 10K records; 95% ≤1s
API response time	≤500 ms (95th percentile)	APM per endpoint	API test suite; 95% ≤500ms
Bulk import (10K records)	≤5 minutes	Import log timestamp	Test with 10K student records
Report generation (complex)	≤30 seconds	Report job log	Generate annual financial report
Image upload (5MB)	≤10 seconds	Upload time tracking	Test from mobile + web
Real-time notification delivery	≤5 seconds (push), ≤30 seconds (SMS)	Notification log vs delivery log	Send test notification 100 users

5.2 Availability

Requirement	Target	Measurement	Acceptance Method
System uptime	≥99.9% monthly (≤43 min downtime)	External uptime monitor (Pingdom)	Monthly uptime report
Planned maintenance window	≤2 hours/month, announced 7 days prior	Maintenance schedule + comms	Verify announcement + actual downtime
Automatic recovery (failure)	≤5 minutes (auto-restart)	Health check + recovery log	Chaos engineering test
Database failover	≤2 minutes	Failover drill log	Quarterly failover drill
RTO (Recovery Time Objective)	≤4 hours	DR drill report	Annual DR drill
RPO (Recovery Point Objective)	≤15 minutes	Backup log + PITR test	Point-in-time recovery test
Mobile app crash-free sessions	≥99.5%	Crashlytics / Sentry	30-day rolling average
API success rate (critical endpoints)	≥99.9%	APM error rate	Monthly API health report

5.3 Scalability

Requirement	Target	Measurement	Acceptance Method
Total schools	10,000+ schools	Tenant count + growth curve	Capacity plan + load test
Branches per school	100+ branches per school	Multi-branch load test	Test with 100-branch single tenant
Concurrent users	100,000+ concurrent	Load test with realistic mix	K6/Gatling test 100K users
Records per tenant	1M+ students, 10M+ audit logs	Database capacity test	Test with seeded data
Horizontal scaling	Auto-scale within 2 minutes	Auto-scaling policy + monitor	Trigger load spike + verify scale

Requirement	Target	Measurement	Acceptance Method
Database sharding readiness	Shard by tenant_id, future-ready	Architecture review	Design review sign-off
File storage scale	1 PB+ object storage	S3 capacity plan	Storage growth projection
Cache hit ratio	≥85% for hot paths	Redis stats	Production monitoring 30 days

5.4 Security

Security NFRs DPDP Act 2023, ISO 27001, SOC 2 Type II compliance चा baseline define करतात. हे requirements PreOne च्या multi-tenant architecture मध्ये tenant isolation ensure करतात आणि PII protection guarantee करतात. Security design review चा framework हा NFR set आहे.

Requirement	Target	Measurement	Acceptance Method
Authentication	JWT + refresh token, MFA for admin	Auth log + MFA enrollment rate	Pen test + auth flow audit
Authorization	RBAC per API endpoint	Permission matrix test	Automated RBAC test suite
Encryption at-rest	AES-256 for database + S3	Config audit	Infrastructure audit
Encryption in-transit	TLS 1.3 mandatory	SSL Labs test	Quarterly SSL scan A+ rating
Tenant isolation	Row-level security, school_id discriminator	Isolation test (cross-tenant access attempt)	Pen test: try cross-tenant access
Audit trail	100% state-changing actions logged	Audit log coverage report	Automated audit coverage test
PII handling	DPDP Act 2023 compliant (consent, retention, DSAR)	DSAR response time + retention audit	DPDP compliance audit
Vulnerability scan	0 critical, ≤2 high (≤30 days to fix)	Quarterly Snyk + Nessus scan	Scan report + remediation SLA

Requirement	Target	Measurement	Acceptance Method
Penetration test	Annual third-party pen test	Pen test report	Remediation of all critical findings
Secure file upload	Type + size validation, virus scan, sandbox storage	Upload attempt log	Test malicious uploads

5.5 Reliability

Requirement	Target	Measurement	Acceptance Method
Retry policy	3 retries with exponential backoff	Retry log + success rate	Inject failures + verify retry
Idempotent APIs	All payment + write APIs idempotent	Idempotency key test	Send duplicate request, verify single effect
Graceful failure	User-friendly error, no PII in message	Error log audit	Force errors + verify UX
Circuit breaker	Trip after 5 failures, half-open after 30s	Circuit breaker state log	Inject downstream failure
Backup & restore	Daily full + continuous WAL	Backup log + restore test	Quarterly restore drill
Database replication	Read replica + sync standby	Replication lag monitor	Verify replication lag $\leq 1s$
Queue durability	Persistent queues with DLQ	Queue depth + DLQ count	Inject poison message + verify DLQ
Health check	Liveness + readiness probes	K8s health endpoint	Manual + automated probe test

5.6 Maintainability

Requirement	Target	Measurement	Acceptance Method
Modular architecture	DDD bounded contexts, ≤7 dependencies per module	Module dependency graph	Architecture review
Clean architecture	Separation of concerns, dependency inversion	Code review + lint	Quarterly architecture audit
API versioning	URI versioning (/v1, /v2), backward compatibility	API spec + breaking change log	API contract test
Code coverage	≥80% line coverage for critical paths	Coverage report (Istanbul/nyc)	CI gate
Structured logging	JSON logs with correlation ID	Log sample + Loki/ELK query	Verify searchability in log tool
Documentation	OpenAPI for all APIs, ADRs for decisions	Doc coverage report	Onboarding test: new dev finds docs
Deployment frequency	Daily deploys possible	Deploy frequency dashboard	CI/CD pipeline uptime
MTTR (Mean Time To Recovery)	≤1 hour for P1 incidents	Incident log	Quarterly incident review

5.7 Usability

Requirement	Target	Measurement	Acceptance Method
Mobile-first	All primary tasks completable on mobile	Task completion test on mobile	Usability test with 10 users
Responsive web	Desktop + tablet + mobile breakpoints	BrowserStack test	Test on 20 device+browser combos

Requirement	Target	Measurement	Acceptance Method
Accessibility	WCAG 2.1 AA compliant	Axe + Lighthouse audit	Accessibility scan score ≥ 90
Task completion time	Attendance: ≤ 2 min/class; Daily sheet: ≤ 3 min/child	Time-on-task measurement	Usability test
Error recovery	Clear error + recovery path	Usability test	Force errors, measure recovery
Onboarding time	First login to first task: ≤ 5 min	Onboarding analytics	New user funnel analysis
Help & documentation	In-app contextual help + searchable docs	Help usage analytics	Search success rate $\geq 80\%$
Visual consistency	Design system adherence $\geq 95\%$	Design lint	Component library audit

5.8 Localization

Requirement	Target	Measurement	Acceptance Method
Languages (v1.0)	English, Marathi, Hindi	Locale coverage	Translation QA per locale
Languages (year 3)	Tamil, Telugu, Kannada, Bengali, Gujarati (8 total)	Locale roadmap	Quarterly locale rollout
i18n architecture	ICU MessageFormat, ready for new locales	Code audit	Add new locale in < 1 day
Date/time format	Per-locale format, IST timezone default	Format test	Verify across locales
Number/currency format	Indian numbering (lakh/crore), ₹ symbol	Format test	Verify INR + Indian format
RTL support	Architecture ready (no RTL in v1.0)	Code review	Design review sign-off

Requirement	Target	Measurement	Acceptance Method
Content translation	UI + system messages + email templates	Translation coverage	Coverage report per locale
Dynamic content	User-generated content stays in original language	Content review	Verify parent posts not auto-translated

5.9 NFR Template

प्रत्येक NFR खालील template नुसार structured आहे — Category, Requirement, Target, Measurement, Acceptance Method. हा template ensure करतो की NFR measurable आणि verifiable आहे:

Field	Description	Example
Category	NFR category (Performance, Security, etc.)	Performance
Requirement	What aspect of quality	Login response time
Target	Measurable target value	≤2 seconds (95th percentile)
Measurement	How it will be measured	APM monitoring + synthetic tests
Acceptance Method	How QA will verify	Load test 1000 concurrent users; 95% ≤2s

6. Business Rules

Business Rules हे PreOne च्या प्रत्येक domain मधील decision points आहेत जे system automatically enforce करते. हे rules BRC v1.0 (Business Rules Catalog) मध्ये comprehensively documented आहेत — 176 rules across 12 categories with 14-field schema per rule. हा section BRC चा executive summary प्रदान करतो आणि key rules example म्हणून दर्शवतो.

Business Rules चा primary purpose हा ensure करणे आहे की system decisions consistent, auditable, आणि compliant आहेत. प्रत्येक rule चा unique ID (R-CATEGORY-XXX format), trigger, condition, action, exception, आणि compliance framework mapping documented आहे. हे rules

engine configuration द्वारा enforce होतात — नवीन rule addition किंवा modification साठी Settings > Workflow Builder वापरा.

6.1 Rule Categories Summary

#	Category	Rules	Key Examples
1	Eligibility & Admission	15	Age eligibility per program, capacity check, duplicate admission prevention, transfer rules
2	Financial & Fee	20	Receipt uniqueness, GST breakdown, refund approval matrix, fee default escalation, scholarship validation
3	Operational	20	One attendance per student per day, daily sheet cutoff, incident severity escalation, meal plan enforcement
4	Academic & Observation	18	Milestone age-appropriate, observation validation, portfolio compilation, report card approval chain
5	HR & Staff	12	Leave balance validation, payroll cutoff, biometric exception, performance review calendar

#	Category	Rules	Key Examples
6	Inventory & Procurement	12	Negative stock prevention, reorder triggers, GRN-PO match, vendor onboarding, audit cadence
7	Communication	12	Parent-child data access, pickup authorization, broadcast audience validation, response SLA
8	Compliance	18	DPDP consent, POSH committee mandate, RTE 25% reservation, Fire NOC validity, GST filing
9	Approval Workflow	15	Multi-level approvals, SLA enforcement, escalation matrix, delegation rules, audit trail
10	Notification	12	Channel preference, opt-out respect, frequency capping, critical bypass, delivery retry
11	Data Governance	12	PII access logging, retention enforcement, DSAR response, anonymization, backup encryption

#	Category	Rules	Key Examples
12	Platform & Multi-tenant	10	Tenant isolation, subscription validity, feature flag enforcement, cross-tenant data prevention

6.2 Sample Business Rules

खालील examples BRC v1.0 मधील representative rules दर्शवतात. प्रत्येक rule चा complete 14-field schema BRC document मध्ये available आहे. येथे फक्त title, trigger, action, आणि compliance mapping दर्शवले आहे:

6.2.1 Eligibility & Admission

Rule ID	Title	Trigger	Action	Compliance
R-ELG-001	Age Eligibility Check	Admission form submission	Reject if age not in program range (1.5-6 years)	ECCE NEP 2020
R-ELG-002	Class Capacity Enforcement	Allocation attempt	Block allocation beyond configured capacity	Internal Policy
R-ELG-003	Duplicate Admission Prevention	New admission record creation	Check phone+aadhaar+name; prompt merge if exists	DPDP §4
R-ELG-004	Document Verification Mandatory	Admission stage advance	Block enrollment until all mandatory docs verified	Maharashtra State Rules

6.2.2 Financial & Fee

Rule ID	Title	Trigger	Action	Compliance
R-FIN-001	Receipt Uniqueness	Receipt generation	Sequential unique number per school per FY	GST Act
R-FIN-002	GST Breakdown Calculation	Invoice generation	Item-wise CGST+SGST @ 18% on taxable services	GST Act §12
R-FIN-003	Refund Approval Matrix	Refund request submission	Principal approval ≤₹5000; Owner approval >₹5000	Internal Policy
R-FIN-004	Fee Default Escalation	Due date passed + unpaid	Day 7: reminder; Day 14: principal alert; Day 30: hold	Internal Policy
R-FIN-005	Negative Stock Prevention	Stock issue attempt	Block issue if stock < quantity requested	Internal Policy

6.2.3 Compliance

Rule ID	Title	Trigger	Action	Compliance
R-CMP-001	DPDP Consent Capture	PII collection (parent/child)	Mandatory consent checkbox + version log	DPDP Act 2023 §4-7
R-CMP-002	POSH Committee Mandate	Staff count ≥10	Alert if ICC not configured; block hiring	POSH Act 2013 §4

Rule ID	Title	Trigger	Action	Compliance
R-CMP-003	RTE 25% Reservation	Admission to entry class	Track 25% seats for EWS/disadvantaged	RTE Act 2009 §12
R-CMP-004	Fire NOC Validity	NOC expiry approaching (60 days)	Alert principal + owner; renewal task	NBC 2016 Fire
R-CMP-005	Data Retention Enforcement	Record retention period reached	Auto-archive alumni after 5 years; purge per DPDP	DPDP §8

6.3 Compliance Framework Mapping

PreOne 8 compliance frameworks चे enforcement ensure करते. प्रत्येक framework चे specific sections PreOne च्या business rules मध्ये mapped आहेत. हे mapping BRC v1.0 Appendix B मध्ये complete detail मध्ये available आहे:

Framework	Scope	Rules Count	Key Sections
DPDP Act 2023	Data protection, consent, retention, DSAR	22	§4 (consent), §8 (retention), §11 (children data), §17 (DSAR), §21 (breach), §33 (transfer)
POSH Act 2013	Workplace sexual harassment prevention	8	§4 (ICC mandate), §9 (complaint), §19 (awareness)
RTE Act 2009	Right to education, 25% reservation	6	§12 (25% EWS reservation), §11 (pre-school)
GST & Finance	Tax compliance, invoicing, TDS	18	GST §12 (services), §31 (invoice), TDS §194C
Fire & Safety (NBC 2016)	Building safety, fire NOC	5	Part 4 Fire, Part 3 Building

Framework	Scope	Rules Count	Key Sections
ECCE NEP 2020	Early childhood curriculum, age criteria	9	Curriculum standards, age 3-6 for entry
ISO 27001	Information security management	14	A.5-A.18 controls
Maharashtra State Preschool Rules	State-specific preschool regulations	10	License, infrastructure, staff qualifications

7. User Roles & Permissions

PreOne चा Role-Based Access Control (RBAC) model 8 primary roles define करतो आणि प्रत्येक role ची per-module permissions specify करतो. हे model ADR-007 (Identity & Authorization) मध्ये freeze केले गेले आहे आणि ते multi-tenant architecture चा core security mechanism आहे.

RBAC चा primary principle हा आहे: "least privilege" — प्रत्येक user ला फक्त त्याच्या role ला आवश्यक असलेले permissions मिळतात. Permissions चे चार levels आहेत: Read (view only), Create (add new), Update (edit existing), Delete (remove). काही actions (e.g., refund approval, fee plan edit) additional approval workflows द्वारे governed आहेत जे role permission पलीकडे आहेत.

7.1 Role Definitions

Role	Description	Primary Device	Reports To
Platform Admin	PreOne SaaS platform operator (internal)	Desktop	PreOne CTO
School Owner	Business owner / investor of preschool	Desktop + Mobile	Self
Principal	Academic + operational head of school	Desktop + Mobile	Owner
Coordinator	Multi-class supervisor / academic coordinator	Tablet + Desktop	Principal

Role	Description	Primary Device	Reports To
Teacher	Classroom teacher / caregiver	Tablet (primary)	Coordinator
Accounts	Finance operations staff	Desktop	Principal / Owner
Reception	Front desk / admissions desk	Desktop + Tablet	Coordinator
Parent	Child's guardian (app user)	Mobile (primary)	N/A

7.2 RBAC Permission Matrix

खालील matrix role × module चे permission level दर्शवते. Permission codes: R = Read, C = Create, U = Update, D = Delete, Full = CRUD all, Own = only own data, Class = only assigned class data. या matrix चा enforcement every API endpoint वर होतो.

Module	Owner	Principal	Coordinator	Teacher	Accounts	Reception	Parent
Marketing & CRM	Full	Full	Full	R Own	R	Full	—
Admissions	R	Full	Full	—	R invoices	Full	R Own + C
Student Lifecycle	R	Full	Full	R Class	R	R	R Own
Academics	R	Full	Full	C+U Class	—	—	R Own child
Daily Operations	R	Full	Full	C+U Class	—	—	R Own child
Parent Communication	R	Full	Full	C+U Class	—	R	C+U Own
Finance	Full	R	—	—	Full	C payment s	R+Pay Own

Module	Owner	Principal	Coordinator	Teacher	Accounts	Reception	Parent
Inventory	R	Full	Full	C request	R+Approve	R	—
HR	R	Full	R	R Own	R payroll	—	—
Administration	R	Full	Full	R	R	C visitors	R
Reports & Analytics	Full	Full	Class	Class	Finance	—	Own child
Settings	R	Full	R	—	R finance	—	Own prefs
Platform Management	—	—	—	—	—	—	—

7.3 Permission Principles

RBAC implementation खालील principles follow करतो. हे principles design review चा framework आहेत — जेव्हा permission ambiguity येतो तेव्हा हे principles tie-breaker म्हणून वापरले जातात:

- **Least Privilege:** Default deny; explicit allow. User ला फक्त त्याच्या role ला आवश्यक असलेले minimum permissions.
- **Need-to-Know:** Data access सीमित — teacher ला फक्त त्याच्या class चा data, parent ला फक्त त्याच्या child चा data.
- **Separation of Duties:** Critical actions ला दोन वेगळे roles — e.g., fee refund request ≠ refund approval (different persons).
- **Approval Workflow:** Role permission पलीकडे high-value actions (refund, salary edit, fee plan change) ला additional approval chain.
- **Audit Every Action:** Permission check result (allow/deny) चा audit log; denial चा reason captured.
- **Context-Aware:** Time, location, device, IP — contextual factors role permission ला override करू शकतात (e.g., after-hours access alert).

- **Temporary Elevation:** Emergency elevation mechanism with auto-expiry + audit + manager notification.
- **Delegation:** Approver out-of-office असल्यास delegation allowed with original approver notified + audit.

8. Module Requirements

हा section PreOne च्या प्रत्येक 13 domain चे detailed module requirements प्रदान करतो. प्रत्येक domain चे overview, goals, business requirements, functional requirements, user stories, business rules, workflows, field specifications, validations, notifications, reports, permissions, APIs, acceptance criteria, आणि estimated size documented आहे.

हे domains DDD bounded contexts म्हणून design केले आहेत — प्रत्येक domain ची स्वतःची data model, business rules, workflows आणि APIs आहेत. Cross-domain communication event-driven architecture द्वारे होते. प्रत्येक domain section त्या domain चा complete specification provide करतो जो engineering team ला implementation करण्यासाठी पुरेसा आहे.

8.1 Marketing & CRM

Marketing & CRM domain preschool चे lead-to-enquiry pipeline manage करतो. यात lead capture (walk-in, call, online, referral, campaign), counselling session booking, follow-up tracking, conversion analytics, आणि campaign ROI tracking यांचा समावेश आहे. Domain चा primary objective म्हणजे every lead ला timely response देणे आणि conversion funnel ची visibility provide करणे.

Goals	Lead response ≤ 30 min; Lead-to-enrolled conversion $\geq 25\%$; Source-wise ROI visibility; Zero lead leakage
Business Requirements	BR-007 Lead Capture & CRM (P0); BR-026 Parent Feedback (P1)
Functional Requirements	FR-001 Lead Capture; FR-002 Follow-up Tracking; FR-003 Counselling Booking; FR-004 Conversion Analytics; FR-005 Campaign Tracking
User Stories	As Reception, I want to capture lead details so I can follow up. As Owner, I want to see conversion funnel so I can optimize. As Marketing, I want to track campaign ROI so I can allocate budget.

Business Rules	R-MKT-001 Lead response SLA 30 min; R-MKT-002 Duplicate merge within 30 days; R-MKT-003 Counsellor availability check; R-MKT-004 Source attribution mandatory; R-MKT-005 Lead score calculation
Workflows	Lead Capture → Pipeline Stage → Follow-up → Counselling → Application → Enrollment; Campaign → Lead Tagging → Conversion → ROI Report
Field Specifications	Lead: name (text, mandatory), phone (10-digit, mandatory), email (email, optional), child_age (1-6), source (enum: walk-in/call/online/referral/campaign), campaign_id (FK, optional), notes (text, ≤500 chars), status (enum), score (auto-calc)
Validations	Phone format; child age range; source from enum; follow-up date > today; counsellor not double-booked
Notifications	New lead → assigned counsellor (push); Follow-up due → reception (push + email); Lead converted → owner + principal (dashboard + email); Cold lead → coordinator (weekly summary)
Reports	Daily lead summary; Weekly conversion funnel; Monthly source-wise ROI; Quarterly campaign performance; Lead aging report
Permissions	Reception: Full; Coordinator: Full; Principal: Full; Owner: Full + analytics; Marketing: Full + campaigns; Teacher: R Own; Parent: —
APIs (Reference)	POST /v1/leads; GET /v1/leads/{id}; PUT /v1/leads/{id}/status; GET /v1/leads/analytics/funnel; POST /v1/campaigns; GET /v1/campaigns/{id}/roi
Acceptance Criteria	AC-001 Lead capture; AC-002 Duplicate handling; AC-003 Conversion report; AC-004 Campaign tracking
Estimated Size	Approx. 35-45 pages detailed spec; ~8,000 LoC; 12 API endpoints; 8 database tables

8.2 Admissions

Admissions domain complete digital admission lifecycle manage करतो — online application form, document upload & verification, multi-stage workflow (application → verification → interaction → offer → enrollment), offer letter generation, आणि student record creation across connected systems. हा domain preschool चा primary revenue driver आहे.

Goals	Admission TAT ≤7 days; 100% document verification; Zero data loss on enrollment; Parent experience smooth
Business Requirements	BR-001 Student Admission (P0); BR-021 Refund Processing (P1)
Functional Requirements	FR-006 Online Application; FR-007 Document Verification; FR-008 Admission Workflow; FR-009 Offer Letter; FR-010 Student Record Creation
User Stories	As Parent, I want to apply online so I don't visit office. As Reception, I want to verify documents digitally. As Principal, I want to track admission stages. As System, I want to create records across all modules on enrollment.
Business Rules	R-ADM-001 Age eligibility per program; R-ADM-002 Capacity enforcement; R-ADM-003 Duplicate prevention; R-ADM-004 Document verification mandatory; R-ADM-005 Offer expiry 7-15 days; R-ADM-006 Sequential stage progression
Workflows	Application → Verification → Interaction → Offer → Acceptance → Fee Payment → Enrollment → Cross-domain Sync; Document Review → Approve/Reject → Re-upload if Rejected
Field Specifications	Application: child_name (mandatory), DOB (date), gender (enum), parent_name (mandatory), parent_phone (10-digit), parent_email (email), address (text), previous_school (optional), medical_info (text), documents (file array), application_fee (auto), status (enum)

Validations	Age 1.5-6 years; phone 10-digit; document formats PDF/JPG ≤5MB; mandatory fields; sequential stages
Notifications	Application submitted → school + parent; Document verified/rejected → parent; Offer letter → parent; Enrollment → all domains + parent welcome
Reports	Daily admission summary; Weekly stage-wise pipeline; Monthly conversion report; Document pending report; Admission source attribution
Permissions	Reception: Full; Principal: Full + approve; Coordinator: Full; Parent: C+R Own; Accounts: R invoices; Owner: R + analytics
APIs (Reference)	POST /v1/applications; GET /v1/applications/{id}; PUT /v1/applications/{id}/verify; POST /v1/applications/{id}/offer; POST /v1/applications/{id}/enroll
Acceptance Criteria	AC-005 Online application; AC-006 Document verification; AC-007 Offer generation; AC-008 Enrollment sync
Estimated Size	Approx. 50-60 pages detailed spec; ~12,000 LoC; 15 API endpoints; 12 database tables

8.3 Student Lifecycle

Student Lifecycle domain child चा complete journey manage करतो — admission पासून graduation पर्यंत. यात class/section allocation, transfers (between classes/sections/branches), promotion to next academic year, archival, alumni access, आणि data retention per DPDP Act यांचा समावेश आहे.

Goals	Zero data loss on transfer; Smooth promotion cycle; DPDP-compliant archival; Alumni access retained 5 years
Business Requirements	BR-008 Student Lifecycle (P0)

Functional Requirements	FR-011 Class & Section Allocation; FR-012 Student Transfer; FR-013 Student Promotion; FR-014 Archives & Alumni
User Stories	As Principal, I want to allocate students to classes optimally. As Coordinator, I want to transfer students between sections. As System, I want to promote students with auto-class allocation. As Alumni, I want to access historical records.
Business Rules	R-STU-001 Age-program match; R-STU-002 Capacity check; R-STU-003 Transfer approval chain; R-STU-004 Promotion eligibility; R-STU-005 Archive after graduation; R-STU-006 Retention 5-7 years
Workflows	Allocation → Eligibility Check → Capacity Check → Assign; Transfer → Approval → Migration → Archive Old; Promotion → Bulk → Eligibility → New Year Record → Class Allocation
Field Specifications	Student: student_id (UUID), name (mandatory), DOB, gender, class_id (FK), section_id (FK), academic_year_id (FK), enrollment_date, status (enum: active/transferred/graduated/archived), alumni_until (date)
Validations	Age matches program; class has capacity; transfer date future; promotion eligibility (completed session)
Notifications	Allocation → parent + teacher; Transfer → both classes' teachers + parent; Promotion → parent + new teacher; Archive → parent notification
Reports	Active students by class/section; Transfer history; Promotion summary; Alumni count; Retention compliance
Permissions	Principal: Full; Coordinator: Full; Teacher: R Class; Parent: R Own child; Owner: R + analytics

APIs (Reference)	POST /v1/students/{id}/allocate; POST /v1/students/{id}/transfer; POST /v1/students/promote; GET /v1/students/{id}/portfolio
Acceptance Criteria	AC-009 Class allocation; AC-010 Student transfer; AC-011 Promotion cycle; AC-012 Alumni access
Estimated Size	Approx. 30-40 pages detailed spec; ~6,000 LoC; 10 API endpoints; 8 database tables

8.4 Academics

Academics domain child च्या learning journey manage करतो — curriculum management, milestone-based observations, portfolio generation, holistic report cards. हा domain preschool चा academic heart आहे जिथे teacher ची expertise आणि AI assistance मिळून child च्या growth ची qualitative record तयार होते.

Goals	Observation frequency ≥ 3 /child/week; 100% milestone tracking; AI suggestion acceptance $\geq 40\%$; Report card generation ≤ 2 hours/class
Business Requirements	BR-009 Academic Observations (P0); BR-020 Report Cards (P1)
Functional Requirements	FR-015 Curriculum Management; FR-016 Observation Recording; FR-017 Milestone Tracker; FR-018 Portfolio Generation; FR-019 Report Card Generation
User Stories	As Teacher, I want to record observations with AI assistance. As Coordinator, I want to see milestone progress across class. As Parent, I want to see child's portfolio. As Principal, I want AI-assisted report cards.
Business Rules	R-ACD-001 Milestone age-appropriate; R-ACD-002 Observation validation; R-ACD-003 Portfolio compilation rules; R-ACD-004 Report card approval chain; R-ACD-005 Curriculum version control; R-ACD-006 AI suggestion threshold

Workflows	Curriculum → Activities → Milestones → Publish; Observation → Select Student → Milestone → AI Suggestion → Publish → Parent; Report Card → AI Draft → Teacher Review → Principal Approval → Parent
Field Specifications	Observation: student_id (FK), milestone_id (FK), text (≥20 chars), photos (file array), activity_context (text), ai_suggestion (text), teacher_approved (bool), published_at (timestamp)
Validations	Observation text ≥20 chars; milestone valid; photo ≤5MB each, max 10; milestone age-appropriate; teacher approval before publish
Notifications	New observation → parent (push + timeline); Milestone achieved → parent + coordinator; Report card ready → parent; Delayed milestone → coordinator alert
Reports	Weekly observation summary; Milestone progress per child/class; Portfolio compilation; Term report card; Curriculum adherence
Permissions	Teacher: C+U Class; Coordinator: Full; Principal: Full + approve; Parent: R Own child; Owner: R + analytics
APIs (Reference)	POST /v1/observations; GET /v1/students/{id}/milestones; POST /v1/students/{id}/portfolio; POST /v1/report-cards; GET /v1/curriculum
Acceptance Criteria	AC-013 Observation recording; AC-014 Milestone tracking; AC-015 Portfolio generation; AC-016 Report card workflow
Estimated Size	Approx. 55-70 pages detailed spec; ~15,000 LoC; 18 API endpoints; 14 database tables + AI integration

8.5 Daily Operations

Daily Operations domain preschool च्या दररोज classroom activities capture करतो — attendance, daily sheet, meals, nap, bathroom, activities, incidents. हा domain teacher चा

primary workspace आहे जिथे ती प्रत्येक child चा दिवस record करते आणि parents ला real-time visibility देते.

Goals	Daily sheet completion $\geq 95\%$; Paper usage reduction 90%+; Real-time parent visibility; Incident response ≤ 5 min
Business Requirements	BR-003 Daily Operations Digital (P0)
Functional Requirements	FR-020 Attendance; FR-021 Daily Sheet; FR-022 Meal Tracking; FR-023 Activity Logging; FR-024 Incident Reporting; FR-025 Nap & Bathroom
User Stories	As Teacher, I want to mark attendance in < 2 min. As Teacher, I want to create daily sheet per child. As Parent, I want to see child's day in real-time. As Principal, I want incident alerts instantly.
Business Rules	R-OPS-001 One attendance per student per day; R-OPS-002 Daily sheet cutoff (end of day); R-OPS-003 Incident severity escalation; R-OPS-004 Meal plan enforcement; R-OPS-005 Activity photo limit; R-OPS-006 Nap duration validation
Workflows	Attendance $\rightarrow$ Open Class $\rightarrow$ Mark $\rightarrow$ Submit $\rightarrow$ Parent Notify; Daily Sheet $\rightarrow$ Pre-populate $\rightarrow$ Fill $\rightarrow$ AI Draft $\rightarrow$ Submit $\rightarrow$ Parent; Incident $\rightarrow$ Capture $\rightarrow$ Severity $\rightarrow$ Instant Alert $\rightarrow$ Follow-up
Field Specifications	Attendance: student_id, date, status (present/absent/late/half-day/leave), marked_by, marked_at, reason_if_absent. Daily Sheet: meals (object), nap (duration), activities (array), mood (emoji), bathroom (count), notes (text)
Validations	One record per student per day; status from enum; submission before cutoff; incident severity from enum; photo limits
Notifications	Absent $\rightarrow$ parent (push + SMS); Late $\rightarrow$ parent; Daily sheet published $\rightarrow$ parent; Critical incident $\rightarrow$ instant push + call prompt; Low meal consumption $\rightarrow$ coordinator

Reports	Daily attendance summary; Daily sheet completion rate; Meal consumption analytics; Incident log; Activity participation
Permissions	Teacher: C+U Class; Coordinator: Full; Principal: Full; Parent: R Own child; Reception: R; Owner: R + analytics
APIs (Reference)	POST /v1/attendance; POST /v1/daily-sheets; POST /v1/meals; POST /v1/activities; POST /v1/incidents; GET /v1/students/{id}/timeline
Acceptance Criteria	AC-017 Attendance; AC-018 Daily sheet; AC-019 Meal tracking; AC-020 Incident reporting; AC-021 Activity logging
Estimated Size	Approx. 60-75 pages detailed spec; ~18,000 LoC; 22 API endpoints; 16 database tables

8.6 Parent Communication

Parent Communication domain school आणि parent दरम्यान real-time, two-way communication enable करतो. यात parent app timeline, two-way chat, announcements/broadcasts, event calendar with RSVP, photo gallery, आणि multi-channel notification delivery यांचा समावेश आहे.

Goals	Parent DAU/MAU $\geq 60\%$; Message response time ≤ 4 hours; Notification delivery $\geq 99\%$; Opt-out rate $\leq 2\%$
Business Requirements	BR-002 Parent Communication (P0); BR-019 Event Management (P1); BR-026 Parent Feedback (P1)
Functional Requirements	FR-026 Parent App Timeline; FR-027 Two-way Chat; FR-028 Announcements; FR-029 Event Calendar & RSVP
User Stories	As Parent, I want to see child's day in timeline. As Parent, I want to chat with teacher. As Principal, I want to broadcast announcements. As Parent, I want to RSVP for events.

Business Rules	R-COM-001 Parent-child data access; R-COM-002 Pickup authorization; R-COM-003 Broadcast audience validation; R-COM-004 Response SLA; R-COM-005 Channel preference; R-COM-006 Frequency capping
Workflows	Timeline → Aggregated from Daily Ops + Academics → Filtered by Child → Parent View; Chat → Send → Notification → Reply → Read Receipt; Announcement → Compose → Audience → Multi-channel → Track Ack
Field Specifications	Message: sender_id, recipient_id, text (≤2000 chars), media (file array), read_at (timestamp), thread_id. Announcement: title, body, audience (school/class/section), attachments, ack_required, channels (array)
Validations	Recipient is child's teacher/parent; message length ≤2000; media ≤5MB; audience scope valid; ack required if mandated
Notifications	New message → push; Announcement → multi-channel per preference; Event reminder → 24h + 1h before; RSVP confirmation → coordinator
Reports	Parent engagement metrics; Message response time; Announcement ack rate; Event RSVP rate; Channel effectiveness
Permissions	Parent: C+U Own; Teacher: C+U Class; Coordinator: Full + broadcast; Principal: Full + broadcast; Owner: R + analytics
APIs (Reference)	GET /v1/parent/timeline/{child_id}; POST /v1/chat/messages; POST /v1/announcements; POST /v1/events; POST /v1/events/{id}/rsvp
Acceptance Criteria	AC-022 Parent timeline; AC-023 Two-way chat; AC-024 Announcement broadcast; AC-025 Event RSVP
Estimated Size	Approx. 40-50 pages detailed spec; ~10,000 LoC; 14 API endpoints; 10 database tables + realtime infra

8.7 Finance

Finance domain preschool चा complete financial operations manage करतो — fee plan configuration, invoice generation, payment processing (UPI/cards/netbanking/cash/cheque), receipt generation, refund processing, expense tracking, आणि financial reporting. हा domain preschool चा revenue engine आहे.

Goals	Fee collection $\geq 92\%$; Online payment adoption $\geq 70\%$; 100% GST compliance; Refund TAT ≤ 7 days
Business Requirements	BR-004 Fee Management (P0); BR-021 Refund Processing (P1)
Functional Requirements	FR-030 Fee Plan Config; FR-031 Invoice Generation; FR-032 Payment Processing; FR-033 Receipt Generation; FR-034 Refund Processing; FR-035 Expense Tracking
User Stories	As Accounts, I want to generate invoices in bulk. As Parent, I want to pay fees online. As System, I want to generate GST-compliant receipts. As Principal, I want to track fee defaults. As Owner, I want expense analytics.
Business Rules	R-FIN-001 Receipt uniqueness; R-FIN-002 GST breakdown; R-FIN-003 Refund approval matrix; R-FIN-004 Fee default escalation; R-FIN-005 Scholarship validation; R-FIN-006 Invoice sequential numbering; R-FIN-007 Reconciliation mandatory
Workflows	Plan $\rightarrow$ Components $\rightarrow$ Publish; Cycle $\rightarrow$ Invoice $\rightarrow$ Parent $\rightarrow$ Payment $\rightarrow$ Reconcile $\rightarrow$ Receipt $\rightarrow$ Ledger; Refund $\rightarrow$ Request $\rightarrow$ Approval $\rightarrow$ Gateway $\rightarrow$ Credit Note $\rightarrow$ Audit
Field Specifications	Invoice: number (unique per school per FY), student_id, components (array), subtotal, GST (object: CGST, SGST, IGST), total, due_date, status (enum). Payment: invoice_id, amount, method, gateway_ref, status, paid_at
Validations	Amount > 0 ; GST rates correct per component; payment method valid; gateway ref unique; refund amount $\leq$ original

Notifications	Invoice generated → parent (push + email); Payment due reminder (3 days before, on due date, 7 days after); Payment success → parent + accounts; Refund processed → parent; Default escalation → principal
Reports	Daily fee collection; Monthly fee summary; GST filing report; Defaulters list; Expense breakdown; P&L summary
Permissions	Accounts: Full; Principal: R; Owner: Full + analytics; Parent: R Own + pay; Reception: C payments
APIs (Reference)	POST /v1/fee-plans; POST /v1/invoices; POST /v1/payments; POST /v1/refunds; GET /v1/finance/gst-report
Acceptance Criteria	AC-026 Fee plan; AC-027 Invoice generation; AC-028 Payment processing; AC-029 Receipt generation; AC-030 Refund workflow
Estimated Size	Approx. 55-65 pages detailed spec; ~14,000 LoC; 20 API endpoints; 14 database tables + payment gateway integration

8.8 Inventory & Procurement

Inventory domain preschool च्या materials चा complete lifecycle manage करतो — purchase requests, purchase orders, goods receipt notes (GRN), stock issues, returns, audits. यात classroom materials (Montessori, art supplies), kitchen stock, office supplies, आणि cleaning materials यांचा समावेश आहे.

Goals	Stock accuracy ≥98%; Wastage reduction 30% +; Reorder automation; 100% audit trail
Business Requirements	BR-005 Inventory Management (P0)
Functional Requirements	FR-036 Purchase Request; FR-037 Purchase Order; FR-038 GRN; FR-039 Stock Issue & Returns

User Stories	As Teacher, I want to request classroom materials. As Coordinator, I want to approve/reject requests. As Accounts, I want to generate POs. As Coordinator, I want to record GRN with quality check. As System, I want to prevent negative stock.
Business Rules	R-INV-001 Negative stock prevention; R-INV-002 Reorder triggers; R-INV-003 GRN-PO match; R-INV-004 Vendor onboarding; R-INV-005 Audit cadence; R-INV-006 FIFO consumption; R-INV-007 Issue purpose mandatory
Workflows	Request → Approval → PO → Vendor → GRN → Stock Update → Invoice Match; Issue → Stock Check → Issue → Ledger → Return Window → Return; Audit → Sample Count → Variance → Reconciliation
Field Specifications	Item: name, category, unit, reorder_level, current_stock, location. PO: vendor_id, line_items (array), taxes, total, payment_terms, delivery_date. GRN: po_id, received_qty, quality_status, remarks
Validations	Quantity > 0; vendor from approved list; received ≤ PO qty; quality check mandatory; issue purpose documented
Notifications	PR submitted → approver; PO issued → vendor + accounts; GRN recorded → accounts; Low stock → coordinator + auto-reorder suggestion; Audit variance → principal
Reports	Stock valuation; Reorder list; Vendor performance; Consumption trends; Audit variance report; Wastage analysis
Permissions	Coordinator: Full; Teacher: C request; Accounts: R + approve; Principal: R; Owner: R + analytics
APIs (Reference)	POST /v1/purchase-requests; POST /v1/purchase-orders; POST /v1/grn; POST /v1/stock-issues; GET /v1/inventory/valuation

Acceptance Criteria	AC-031 Purchase request; AC-032 PO generation; AC-033 GRN; AC-034 Stock issue; AC-035 Negative stock prevention
Estimated Size	Approx. 40-50 pages detailed spec; ~9,000 LoC; 12 API endpoints; 10 database tables

8.9 Human Resources

HR domain preschool च्या staff चा complete lifecycle manage करतो — recruitment, leave management, payroll processing, performance reviews, training, आणि offboarding. हा domain staff ची productivity आणि satisfaction ensure करतो जे child care quality ला directly impact करते.

Goals	Payroll accuracy 100%; Leave transparency 100%; Performance review on-time $\geq 90\%$; Staff retention $\geq 85\%$
Business Requirements	BR-010 HR & Payroll (PO)
Functional Requirements	FR-040 Recruitment; FR-041 Leave Management; FR-042 Payroll; FR-043 Performance Review
User Stories	As HR, I want to manage recruitment pipeline. As Staff, I want to apply for leave online. As Accounts, I want to process payroll accurately. As Coordinator, I want to conduct performance reviews. As Principal, I want staff analytics.
Business Rules	R-HR-001 Leave balance validation; R-HR-002 Payroll cutoff; R-HR-003 Biometric exception; R-HR-004 Performance review calendar; R-HR-005 Recruitment approval; R-HR-006 Probation period; R-HR-007 Exit workflow
Workflows	Recruitment → Posting → Applications → Screening → Interview → Offer → Onboard; Leave → Apply → Balance Check → Approve → Calendar → Coverage; Payroll → Cycle → Attendance → Calc → Payslip → Bank File

Field Specifications	Staff: name, role, doj, salary_structure, bank_details, status. Leave: staff_id, type, dates, reason, status, balance_impact. Payroll: staff_id, month, gross, deductions, net, bank_file_ref
Validations	Leave balance sufficient; payroll cycle complete; salary structure current; bank details valid; rating scale valid
Notifications	Leave applied → approver; Leave approved/rejected → staff; Payslip generated → staff; Performance review due → manager + staff; Payroll processed → accounts + owner
Reports	Staff directory; Leave summary; Payroll register; Performance dashboard; Recruitment pipeline; Attrition analysis
Permissions	Principal: Full; HR: Full; Coordinator: R + leave approve; Staff: R Own + leave apply; Accounts: R payroll; Owner: R + analytics
APIs (Reference)	POST /v1/leaves; POST /v1/payroll/run; GET /v1/staff/{id}/payslip; POST /v1/performance-reviews; GET /v1/hr/analytics
Acceptance Criteria	AC-036 Leave workflow; AC-037 Payroll processing; AC-038 Performance review; AC-039 Recruitment
Estimated Size	Approx. 45-55 pages detailed spec; ~11,000 LoC; 16 API endpoints; 12 database tables + biometric integration

8.10 Administration

Administration domain preschool च्या facility आणि compliance operations manage करतो — asset management, visitor management, maintenance scheduling, compliance dashboard, meal plans. हा domain preschool चा physical operations backbone आहे.

Goals	Asset tracking 100%; Maintenance schedule adherence ≥90%; 100% compliance adherence; Visitor auth 100%
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Business Requirements	BR-011 Compliance Management (P0); BR-017 Visitor Management (P1); BR-018 Asset Management (P1)
Functional Requirements	FR-044 Asset Management; FR-045 Visitor Management; FR-046 Maintenance Scheduling; FR-047 Compliance Dashboard
User Stories	As Coordinator, I want to track all assets. As Reception, I want to digitize visitor check-in. As Principal, I want compliance dashboard. As System, I want to alert before compliance expiry.
Business Rules	R-ADMIN-001 Asset tagging; R-ADMIN-002 Maintenance frequency; R-ADMIN-003 Visitor photo mandatory; R-ADMIN-004 Pickup authorization; R-ADMIN-005 Compliance alert thresholds; R-ADMIN-006 Audit cadence
Workflows	Asset → Entry → Tag → Assign → Maintenance → Audit → Depreciation → Disposal; Visitor → Capture → Photo → Host Notify → Approve → Check-in → Check-out; Compliance → Items → Status → Expiry Alert → Action → Audit
Field Specifications	Asset: name, category, value, purchase_date, location, assigned_to, qr_code, depreciation. Visitor: name, phone, photo, purpose, host, child_id (if pickup), check_in, check_out
Validations	Asset value > 0; visitor phone + photo mandatory; pickup → authorized contact verification; compliance due dates tracked
Notifications	Asset assigned/reassigned → staff; Maintenance due → coordinator + vendor; Visitor check-in → host; Unauthorized pickup → instant deny + parent call; Compliance expiry approaching → principal + owner
Reports	Asset register; Maintenance schedule; Visitor log; Compliance status; Depreciation report; Pickup audit
Permissions	Coordinator: Full; Principal: Full; Reception: C visitors; Accounts: R assets; Owner: R + analytics

APIs (Reference)	POST /v1/assets; POST /v1/visitors; POST /v1/maintenance; GET /v1/compliance/dashboard; GET /v1/admin/audit
Acceptance Criteria	AC-040 Asset tracking; AC-041 Visitor check-in; AC-042 Maintenance; AC-043 Compliance alert
Estimated Size	Approx. 35-45 pages detailed spec; ~8,000 LoC; 14 API endpoints; 10 database tables

8.11 Reports & Analytics

Reports & Analytics domain PreOne चा intelligence layer आहे जो सर्व domains च्या data ला aggregate करून actionable insights generate करतो — operational dashboards, custom report builder, predictive analytics, आणि role-based KPI tracking.

Goals	Dashboard DAU ≥80% management; Decision speed 5x faster; Custom report builder adoption ≥30%; Prediction accuracy ≥75%
Business Requirements	BR-012 Real-time Dashboards (P0)
Functional Requirements	FR-048 Operational Dashboards; FR-049 Custom Report Builder; FR-050 Predictive Analytics
User Stories	As Owner, I want real-time KPI dashboard. As Principal, I want operational alerts. As Coordinator, I want class-level analytics. As Owner, I want to build custom reports. As Principal, I want AI predictions for capacity planning.
Business Rules	R-RPT-001 Role-based data scope; R-RPT-002 Data refresh cadence; R-RPT-003 Export format validation; R-RPT-004 Prediction confidence display; R-RPT-005 Dashboard customization limits
Workflows	Data Sources → Aggregation → Role Filter → Dashboard Render → Drill-down → Export; Custom Report → Source → Fields → Filters → Preview → Save → Schedule

Field Specifications	Dashboard: user_role, widgets (array), filters (object), refresh_interval. Custom Report: data_source, fields (array), filters (object), grouping, schedule (cron)
Validations	Role sees only authorized scope; data refresh ≤5 min; field-data source compatibility; schedule frequency valid
Notifications	Scheduled report ready → email; Threshold breach → push + email; Prediction alert → designated role; Dashboard anomaly → principal
Reports	Daily KPI snapshot; Weekly operational summary; Monthly business review; Quarterly trend analysis; Annual strategic report
Permissions	Owner: Full + analytics; Principal: Full; Coordinator: Class scope; Accounts: Finance scope; Teacher: Class scope; Parent: Own child
APIs (Reference)	GET /v1/dashboards/{role}; POST /v1/reports/custom; GET /v1/reports/{id}/download; POST /v1/analytics/predict
Acceptance Criteria	AC-044 Dashboard load; AC-045 Custom report; AC-046 Predictive analytics
Estimated Size	Approx. 30-40 pages detailed spec; ~7,000 LoC; 10 API endpoints + ML model integration; 6 database views + materialized views

8.12 Settings

Settings domain school च्या configurations चा master control आहे — school profile, academic year, programs, classes, sections, timings, notification rules, workflow builder, integration settings. हा domain बाकीच्या सर्व domains चा behavior customize करतो.

Goals	100% configuration flexibility; 0 breaking changes; Workflow builder adoption ≥30%; Notification rules per school ≥10
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Business Requirements	BR-023 Custom Workflows (P1)
Functional Requirements	FR-051 School Configuration; FR-052 Notification Rules; FR-053 Workflow Builder
User Stories	As Principal, I want to configure school profile. As Coordinator, I want to set notification rules. As Principal, I want to customize approval workflows. As Owner, I want to manage integrations.
Business Rules	R-SET-001 Config versioning; R-SET-002 Cascade impact check; R-SET-003 Workflow guardrails; R-SET-004 Notification template validation; R-SET-005 Integration auth refresh; R-SET-006 Feature flag governance
Workflows	Config Change → Impact Analysis → Confirmation → Apply → Cascade → Audit; Notification Rule → Event → Channel → Template → Audience → Test → Activate
Field Specifications	School Config: name, address, programs (array), classes (array), sections (array), timings (object), academic_years (array). Notification Rule: event, channel, template, audience, timing, active
Validations	Programs age-appropriate; classes capacity > 0; timings valid; channel enabled; template valid; audience scope valid
Notifications	Config change → stakeholders; Notification rule activated → test send; Workflow activated → audit; Integration failure → admin
Reports	Configuration audit log; Notification rule performance; Workflow usage stats; Integration health
Permissions	Principal: Full; Owner: R + analytics; Coordinator: R + notifications; Platform Admin: Full + integrations
APIs (Reference)	PUT /v1/school/config; POST /v1/notification-rules; POST /v1/workflows; GET /v1/integrations/health

Acceptance Criteria	AC-047 School config; AC-048 Notification rules; AC-049 Workflow builder
Estimated Size	Approx. 25-35 pages detailed spec; ~5,000 LoC; 8 API endpoints; 6 database tables + config store

8.13 Platform Management

Platform Management domain PreOne चा SaaS operations layer आहे जो multi-tenant infrastructure manage करतो — tenant onboarding, subscription & billing, audit log & compliance, feature flags, system health. हा domain internal PreOne team चा primary workspace आहे.

Goals	10,000+ schools scale; 99.9% uptime; NRR $\geq 110\%$; 100% audit trail; Subscription churn $\leq 5\%$
Business Requirements	BR-006 Multi-School SaaS (P0); BR-013 Notification Engine (P0); BR-014 Audit Trail (P0); BR-015 Mobile Apps (P0); BR-016 Multi-language (P1); BR-022 Bulk Import (P1)
Functional Requirements	FR-054 Tenant Onboarding; FR-055 Subscription & Billing; FR-056 Audit Log & Compliance
User Stories	As Platform Admin, I want to onboard new tenants. As Platform Admin, I want to manage subscriptions. As Platform Admin, I want to view audit trail. As Platform Admin, I want to monitor system health.
Business Rules	R-PLT-001 Tenant isolation; R-PLT-002 Subscription validity; R-PLT-003 Feature flag enforcement; R-PLT-004 Cross-tenant prevention; R-PLT-005 Audit immutability; R-PLT-006 Retention 7 years
Workflows	Tenant → Create → Config → Admin User → Sample Data → Training → Go-Live; Subscription → Plan → Billing → Invoice → Payment → Renewal; Audit → Capture → Sign → Store → Search → Retain

Field Specifications	Tenant: name, plan, admin_user, status, created_at, go_live_date. Subscription: tenant_id, plan, billing_cycle, amount, status, renewal_date. Audit: actor, action, target, before, after, ip, timestamp
Validations	Unique tenant ID; admin email unique; plan from catalog; audit immutable; retention enforced
Notifications	Tenant onboarded → internal team; Subscription renewal approaching → owner; Payment failure → dunning; Audit anomaly → security team; System health degraded → on-call
Reports	Tenant growth; MRR/ARR; Churn analysis; Audit coverage; System health; Feature adoption
Permissions	Platform Admin: Full; Owner: R Own tenant; Principal: — ; Other roles: —
APIs (Reference)	POST /v1/platform/tenants; POST /v1/platform/subscriptions; GET /v1/platform/audit; GET /v1/platform/health
Acceptance Criteria	AC-050 Tenant onboarding; AC-051 Subscription billing; AC-052 Audit log; AC-053 Multi-tenant isolation
Estimated Size	Approx. 35-45 pages detailed spec; ~9,000 LoC; 12 API endpoints; 8 database tables + multi-tenant infra

9. Reports & Analytics

Reports & Analytics section PreOne च्या reporting capabilities चा comprehensive catalog प्रदान करतो. Reports चे 5 categories मध्ये classified केले आहे — Operational, Academic, Financial, Management, Compliance. प्रत्येक report चा audience, format, frequency, आणि export options documented आहे.

Reports चा primary objective म्हणजे data-driven decision making enable करणे. Real-time dashboards daily operations monitor करतात, scheduled reports periodic reviews support

करतात, आणि on-demand reports ad-hoc analysis enable करतात. सर्व reports role-based access controls follow करतात — user ला फक्त त्याच्या role ला authorized data दिसतो.

9.1 Operational Reports

Report Name	Audience	Format	Frequency	Export
Daily Attendance Summary	Teacher, Coordinator, Principal	Dashboard + PDF	Real-time + Daily 6 PM	PDF, Excel
Daily Sheet Completion	Coordinator, Principal	Dashboard	Real-time	Excel
Weekly Activity Log	Coordinator, Principal	PDF	Weekly Monday	PDF, Excel
Incident Log	Principal, Owner	Dashboard + PDF	Real-time + Daily	PDF
Meal Consumption Report	Coordinator, Kitchen	Dashboard	Real-time + Daily	Excel
Visitor Log	Coordinator, Principal	Dashboard + PDF	Daily	PDF, Excel
Stock Movement Report	Coordinator, Accounts	Dashboard	Real-time + Weekly	Excel
Staff Attendance Summary	Principal, HR	Dashboard	Daily	Excel

9.2 Academic Reports

Report Name	Audience	Format	Frequency	Export
Milestone Progress per Child	Teacher, Parent	Dashboard	Real-time	PDF
Class Milestone Summary	Coordinator, Principal	Dashboard	Weekly	Excel

Report Name	Audience	Format	Frequency	Export
Observation Frequency Report	Coordinator, Principal	Dashboard	Weekly	Excel
Child Portfolio	Parent, Teacher	PDF	On-demand + Term-end	PDF
Report Card (Term)	Parent, Principal	PDF	Term-end	PDF
Curriculum Adherence Report	Coordinator, Principal	Dashboard + PDF	Weekly	PDF, Excel
Delayed Milestone Alert Report	Coordinator, Principal	Dashboard + Alert	Real-time + Weekly	Excel
AI Suggestion Acceptance Report	Principal, Product	Dashboard	Monthly	Excel

9.3 Financial Reports

Report Name	Audience	Format	Frequency	Export
Daily Fee Collection	Accounts, Principal, Owner	Dashboard	Real-time + Daily	Excel
Monthly Fee Summary	Accounts, Owner	PDF + Excel	Monthly	PDF, Excel
Defaulters List	Accounts, Principal	Dashboard + PDF	Real-time + Weekly	PDF, Excel
GST Filing Report	Accounts	Excel + PDF	Monthly + Quarterly	Excel, PDF
Refund Summary	Accounts, Principal	PDF	Monthly	PDF, Excel
Expense Breakdown	Accounts, Owner	Dashboard + PDF	Monthly	Excel

Report Name	Audience	Format	Frequency	Export
P&L Summary	Owner	PDF	Monthly + Quarterly	PDF
Fee Plan Performance	Principal, Owner	Dashboard	Quarterly	Excel

9.4 Management Reports

Report Name	Audience	Format	Frequency	Export
Admission Conversion Funnel	Owner, Principal	Dashboard	Real-time	PDF, Excel
Branch Performance Comparison	Owner	Dashboard	Monthly	Excel
Teacher Productivity Report	Principal, Owner	Dashboard	Monthly	Excel
Parent NPS / CSAT Report	Owner, Principal	Dashboard + PDF	Quarterly	PDF
Staff Retention / Attrition	Principal, Owner	Dashboard	Quarterly	Excel
Enrollment Trend Analysis	Owner	Dashboard + PDF	Monthly	PDF, Excel
Revenue Trend & Forecast	Owner	Dashboard	Monthly	Excel
Capacity Utilization Report	Principal, Owner	Dashboard	Monthly	Excel

9.5 Compliance Reports

Report Name	Audience	Format	Frequency	Export
DPDP Consent Register	Principal, Owner	PDF	On-demand	PDF
DPDP DSAR Response Log	Principal, Owner	PDF	On-demand	PDF
POSH Compliance Report	Principal, Owner	PDF	Annual	PDF
RTE 25% Reservation Tracker	Principal, Owner	Dashboard + PDF	Annual admission cycle	PDF, Excel
Fire & Safety NOC Status	Principal, Owner	Dashboard	Real-time	PDF
GST Reconciliation Report	Accounts	Excel + PDF	Monthly	Excel, PDF
Audit Trail Export	Platform Admin, Principal	Excel + PDF	On-demand	Excel, PDF
Data Retention Compliance	Principal, Owner	PDF	Quarterly	PDF

10. Integrations

PreOne च्या platform वर 8 integration categories आहेत ज्या external systems सोबत communicate करतात. प्रत्येक integration चा provider, scope, auth method, SLA, आणि fallback strategy documented आहे. Integrations loosely coupled architecture द्वारे implement होतात — एक integration fail होतो तर बाकीचे system unaffected राहतो.

Integration design principles: (1) Loose coupling via event bus + queue; (2) Idempotent operations with retry; (3) Fallback chain (primary → secondary → manual); (4) Rate limit handling with backoff; (5) Audit log of every external call; (6) Health check + alerting per integration; (7) Cost tracking per integration; (8) Vendor lock-in prevention via abstraction layer.

10.1 Payment Gateway Integration

Provider	Scope	Auth	SLA	Fallback
Razorpay (Primary)	UPI, cards, netbanking, wallets	API key + secret (HMAC)	99.95% uptime, 2s response	Cashfree automatic
Cashfree (Secondary)	UPI, cards, netbanking	API key + secret	99.9% uptime	Manual entry / cheque
UPI Direct (Reserve)	UPI Collect, Intent	Merchant API	99.5% uptime	Razorpay/ Cashfree
Cash/Cheque (Offline)	Manual entry by reception	N/A — internal record	Real-time entry	N/A

10.2 SMS Gateway Integration

Provider	Scope	Auth	SLA	Fallback
MSG91 (Primary)	Transactional + OTP SMS India	Auth key	99.5% delivery, ≤30s	Twilio automatic
Twilio (Secondary)	International SMS fallback	Account SID + token	99% delivery	Push notification
Gupshup (Reserve)	Promotional SMS (with consent)	API key	99% delivery	WhatsApp

10.3 WhatsApp Business Integration

Provider	Scope	Auth	SLA	Fallback
WhatsApp Cloud API (Meta)	Template messages, media, two-way chat	Bearer token + Webhook verify	Meta SLA, ≤30s delivery	SMS fallback
Interakt (Reserve)	Template + broadcast	API key	99% delivery	Cloud API

10.4 Email Integration

Provider	Scope	Auth	SLA	Fallback
AWS SES (Primary)	Transactional emails (invoices, receipts, notifications)	SMTP / API + IAM	99.9% delivery	SendGrid automatic
SendGrid (Secondary)	Marketing emails + backup	API key	99.5% delivery	Manual via Gmail

10.5 Biometric Integration

Provider	Scope	Auth	SLA	Fallback
eSSL (Primary)	Staff attendance (fingerprint)	Device SDK + local API	Local network	Manual entry
Mantra (Secondary)	Aadhaar-compatible biometric	Device SDK	Local network	eSSL
Face Recognition (Future)	Child + staff touchless attendance	AI model + camera	TBD	Card / biometric

10.6 AI / LLM Integration

Provider	Scope	Auth	SLA	Fallback
Z.ai GLM (Primary)	Observation assistant, daily sheet draft, report card narrative, parent reply composer	API key	99.5% uptime, 3s response	Manual writing

Provider	Scope	Auth	SLA	Fallback
OpenAI (Reserve)	Backup LLM if primary down	API key	99.5% uptime	GLM
Vision Model (For Photos)	Photo tagging, child identification (within school)	API key	99% accuracy	Manual tagging
Speech-to-Text (Future)	Voice daily sheet entry	API key	99% accuracy	Text input

10.7 Cloud Storage Integration

Provider	Scope	Auth	SLA	Fallback
AWS S3 (Primary)	Photos, documents, exports, backups	IAM role + bucket policy	99.99% durability	Multi-region replication
Cloudflare R2 (Secondary)	Cost-effective bulk storage for archives	API token	99.99% durability	S3
Cloudfront CDN	Image delivery to apps	Origin + distribution	99.99% availability	Direct S3

10.8 Identity & SSO Integration

Provider	Scope	Auth	SLA	Fallback
Internal JWT (Primary)	All user authentication	JWT + refresh token	99.99% (internal)	OTP fallback
Google OAuth (Future)	Admin SSO for owners/principals	OAuth 2.0	Google SLA	Internal auth
Microsoft OAuth (Future)	Enterprise SSO	OAuth 2.0	Microsoft SLA	Internal auth

11. Acceptance Criteria (AC)

Acceptance Criteria हे QA Team साठी definitive validation rules आहेत. प्रत्येक AC Given/When/Then format मध्ये structured आहे ज्यामुळे test automation सोपं होतं. हे 50 ACs 13 domains च्या critical flows cover करतात.

AC चा primary purpose हा ensure करणे आहे की implementation acceptance च्या वेळी user requirements meet होतात. प्रत्येक AC चा unique ID (AC-XXX format), scenario description, Given (preconditions), When (trigger action), Then (expected outcome), आणि expected result आहे. QA team या ACs पासून test cases derive करतो.

AC ID	Scenario	Given	When	Then	Expected
AC-001	Lead Capture — Valid	Reception logged in	Capture lead with all mandatory fields	Lead created with unique ID, pipeline entry, counsellor notified	Lead visible in pipeline within 5 seconds; notification delivered
AC-002	Lead Capture — Duplicate	Lead exists with phone 9876543210	New lead entered with same phone within 30 days	System prompts merge with existing lead	Merge option shown; no duplicate created
AC-003	Conversion Report	Leads in various stages	Owner views conversion funnel for last 30 days	Funnel displayed with stage-wise conversion %	Report loads ≤3s; data accurate within 5 min
AC-004	Campaign Tracking	Active campaign with UTM	Leads tagged with campaign	Campaign ROI tracked	Cost-per-lead and cost-per-admission calculated

AC ID	Scenario	Given	When	Then	Expected
AC-005	Online Application — Valid	Parent has app access	Submit application with all mandatory fields + documents	Application ID generated, acknowledgment sent	Application visible to school within 10s; parent notified
AC-006	Document Verification	Application submitted with documents	Reception reviews and approves documents	Workflow advances to next stage; parent notified	[Please fill in]
AC-007	Offer Letter Generation	Application reached offer stage	Principal generates offer letter	PDF generated with merge fields, parent notified	Offer expiry 7-15 days from issue; unique offer ID
AC-008	Enrollment & Cross-domain Sync	Offer accepted + admission fee paid	System creates student record	Records created in all 13 domains; welcome notification sent	Sync complete within 30s; zero data loss
AC-009	Class Allocation — Capacity OK	Class has 18 students, capacity 25	Allocate new student	Allocation successful; roster updated	Teacher notified; parent notified
AC-010	Class Allocation — Capacity Full	Class at 25/25 capacity	Attempt allocation	System blocks + suggests alternative	Waitlist option offered
AC-011	Student Transfer	Student enrolled in Class A	Transfer to Class B (with capacity)	Transfer record created; old archived; new roster updated	Both teachers + parent notified

AC ID	Scenario	Given	When	Then	Expected
AC-012	Promotion Cycle	Academic year ending; students completed session	Principal triggers bulk promotion	All eligible students promoted to next program	New year records created; ineligible flagged for review
AC-013	Observation Recording	Teacher logged in; student in class	Teacher records observation with milestone + text + photo	Observation saved; parent notified; milestone progress updated	AI suggestion offered; teacher can accept/reject/edit
AC-014	Milestone Tracking	Multiple observations recorded for child	View milestone tracker	Progress shown per development area	Delayed milestones flagged with alert
AC-015	Portfolio Generation	Observations recorded over a term	Generate portfolio PDF	PDF compiled with observations + photos + milestones	File size ≤25 MB; parent shareable link generated
AC-016	Report Card Workflow	Term completed; observations recorded	Teacher triggers report card generation	AI draft → teacher review → principal approval → parent share	AI draft ready ≤30s; full cycle ≤2 hours per class
AC-017	Attendance — Valid	Teacher logged in; class roster loaded	Mark attendance for all students + submit	Records saved; parents of absent/late notified	One record per student per day; submission before cutoff
AC-018	Attendance — Duplicate	Attendance already marked for student today	Teacher submits again	System displays duplicate attendance validation	Edit allowed with reason; original preserved in audit

AC ID	Scenario	Given	When	Then	Expected
AC-019	Daily Sheet	Student present today	Teacher fills daily sheet + submits	Sheet published to parent timeline; analytics updated	AI draft assist offered; all sections validated
AC-020	Incident — Critical	Student present	Teacher reports critical incident	Instant parent push + call prompt; coordinator + principal alerted	Notification ≤5s; follow-up task auto-created
AC-021	Activity Logging with Photos	Activity scheduled	Teacher logs activity with photos	Photos uploaded to gallery; parent timeline updated	Photo ≤5MB each, max 10; upload ≤10s
AC-022	Parent Timeline	Child has activities today	Parent opens app	Timeline shows today's events reverse-chronological	Load ≤3s; only own child data visible
AC-023	Two-way Chat	Parent + teacher both have app access	Parent sends message to teacher	Teacher gets push notification; can reply; read receipts	Delivery ≤5s; SLA-based response reminder
AC-024	Announcement Broadcast	Principal composes announcement for school	Send to school audience	Multi-channel delivery (push + email + WhatsApp); ack tracked	Delivery ≥99% per channel; ack rate tracked

AC ID	Scenario	Given	When	Then	Expected
AC-025	Event RSVP	Event created with RSVP	Parent RSVPs yes/no	RSVP recorded; coordinator sees count	Reminder sent 24h + 1h before; photo gallery post-event
AC-026	Fee Plan Creation	Principal logged in; academic year configured	Create fee plan with components + amounts	Plan published; available for enrollment	Validation: amount > 0; components non-overlapping
AC-027	Invoice Generation	Students enrolled with fee plan	Cycle triggers invoice generation	Invoices generated per student; GST breakdown correct	Unique sequential invoice number; due date ≥7 days
AC-028	Payment Success	Invoice generated	Parent pays via UPI successfully	Payment recorded; receipt generated; ledger updated	Receipt unique; parent notified; accounts dashboard updated
AC-029	Receipt Generation	Payment successful	System generates receipt	Receipt PDF with unique number + GST breakdown	Number sequential per school per FY; downloadable + archived
AC-030	Refund Workflow	Payment exists; refund requested	Process refund per approval matrix	Approval → gateway refund → credit note → audit trail	Principal ≤₹5000; Owner >₹5000; TAT ≤7 days
AC-031	Purchase Request	Coordinator logged in; items in catalog	Raise PR with item + qty + justification	PR created; approval workflow triggered	Justification ≥20 chars; expected date ≥3 days from now

AC ID	Scenario	Given	When	Then	Expected
AC-032	PO Generation	PR approved	Generate PO with vendor + line items	PO PDF generated; vendor emailed; acknowledgment tracked	Vendor from approved list; total > 0; terms valid
AC-033	GRN Recording	PO sent to vendor; goods received	Record GRN with quantity + quality check	Stock incremented; invoice matched; payment release triggered	Received ≤ PO qty; quality check mandatory
AC-034	Stock Issue	Stock available	Issue stock to classroom with purpose	Stock decremented; ledger updated; returnable flag set	Stock ≥ issue qty; purpose documented
AC-035	Negative Stock Prevention	Stock at 5 units	Attempt issue of 10 units	System blocks issue; suggests reorder	Error message clear; reorder suggestion shown
AC-036	Leave Workflow	Staff has leave balance	Apply for leave with dates + reason	Approval chain triggered; calendar updated on approval	Balance check; coverage alert if critical role
AC-037	Payroll Processing	Attendance + leave data complete for month	Accounts triggers payroll run	Payslips generated; bank file created; ledger updated	100% accuracy; staff notified; bank file format valid

AC ID	Scenario	Given	When	Then	Expected
AC-038	Performance Review	Staff enrolled ≥3 months	Manager initiates review	Self + peer + manager feedback cycle; final rating + dev plan	All goals rated; feedback ≥50 chars per goal
AC-039	Recruitment	Position approved with budget	HR posts job + collects applications	Pipeline: applications → screening → interview → offer → onboard	Offer within budget; onboarding tasks auto-created
AC-040	Asset Tracking	Asset registered with QR	Scan QR to view asset details	Asset record shown with location, assignment, maintenance	Depreciation auto-calculated; audit trail visible
AC-041	Visitor Check-in	Visitor arrives; reception logged in	Capture details + photo + purpose + host	Host notified; visitor badge printed; check-in logged	Photo mandatory; pickup → child auth contact verified
AC-042	Unauthorized Pickup	Visitor attempts child pickup	Visitor not in authorized contacts list	System denies; parent called; alert raised	Child not released; principal alerted; audit logged
AC-043	Compliance Alert	Compliance item expiry approaching (60 days)	System checks daily	Alert raised to principal + owner; renewal task created	Critical if expired; dashboard reflects status
AC-044	Dashboard Load	Owner logs in	Open dashboard	Real-time KPIs displayed per role	Load ≤3s; data refresh ≤5 min; role-based scope enforced

AC ID	Scenario	Given	When	Then	Expected
AC-045	Custom Report	User has report-builder permission	Build custom report with fields + filters	Report preview + save + schedule options	Fields compatible with source; heavy query async
AC-046	Predictive Analytics	Sufficient historical data (≥ 6 months)	Run prediction for admission conversion	Predictions with confidence score + recommendations	Confidence displayed; recommendation tied to prediction
AC-047	School Config Change	Principal logged in	Edit school config (e.g., add class)	Impact analysis shown; cascade with confirmation	Affects enrolled students → explicit confirmation
AC-048	Notification Rule	Coordinator has settings permission	Create notification rule for event	Rule saved; test send executed; audit logged	Channel enabled; template valid; test delivers
AC-049	Workflow Builder	Principal with workflow builder permission	Customize approval workflow	Within guardrails → saved + activated	Guardrail violation → blocked with explanation
AC-050	Tenant Onboarding	Sales closed; contract signed	Platform admin creates new tenant	Tenant created; admin user; sample data; go-training scheduled	Unique tenant ID; admin email unique; go-live plan

11.1 Acceptance Criteria Template

प्रत्येक AC खालील template नुसार structured आहे. QA team या template चा वापर करून test cases लिहितो आणि automation scripts develop करतो:

Field	Description	Example
Requirement ID	Stable identifier AC-XXX	AC-018
Scenario	Brief scenario description	Attendance already marked
Given	Precondition / state	Attendance already marked for student today
When	Trigger action	Teacher submits again
Then	Expected system behavior	System displays duplicate attendance validation
Expected Result	Measurable outcome	Edit allowed with reason; original preserved in audit

12. Out of Scope

Out of Scope section clearly define करतो काय PreOne v1.0 मध्ये deliver होणार नाही. हे scope management चा critical tool आहे — feature creep prevent करतो आणि v1.0 च्या release timeline ला realistic ठेवतो. खालील items explicitly future versions किंवा partner ecosystem कडे delegated केले आहेत.

Out of scope decisions चा rationale तीन principles वर आधारित आहे: (1) Core focus preservation — preschool-specific workflows ला priority; (2) Build vs Buy — generic capabilities (e.g., accounting) ला integrate करणे अपेक्षित नाही; (3) v1.0 timeline — 12-month delivery window मध्ये feasible scope only. प्रत्येक out-of-scope item चा revisit plan documented आहे.

12.1 Explicitly Excluded from v1.0

Item	Reason	Revisit Plan
K-12 / College / University support	PreOne हे exclusively preschool-specific; K-12 fundamentally different data model	Out of strategic scope permanently
Full accounting system (Tally replacement)	Tally/Busy already mature; integrate via export only	v1.1: Tally XML export; v1.2: real-time sync API
Transport management with live GPS	Complex operational feature; not core to preschool	v1.1+ with partner integration

Item	Reason	Revisit Plan
Hostel / Residential management	Not applicable to preschool segment	Out of strategic scope
Library management	Preschools rarely have formal libraries	v1.2 if demand emerges
Examination & gradebook (K-12 style)	Preschools use observations + milestones, not exams	Out of strategic scope permanently
Multi-country tax compliance	v1.0 India-only (GST); international tax complex	Year 3+ with global expansion
Third-party marketplace	Requires significant seller onboarding infrastructure	v2.0 with ecosystem play
Parent community / social network	Privacy concerns; not core to operations	v2.0 with opt-in model
AI voice-based daily sheet entry	Speech-to-text accuracy for Indian languages still improving	v1.2 with proven accuracy
Biometric child attendance (face recognition)	Privacy concerns with minors; tech not mature	v1.2 with parent consent + legal review
Wearable device integration (smart bands)	Cost-prohibitive for preschools	v2.0+ with partner ecosystem
Multi-currency support	India-only v1.0	Year 3+ with global expansion
Offline-first mobile app	Online-first acceptable for v1.0; offline complex	v1.2 for low-connectivity areas
Custom integrations marketplace	Requires developer ecosystem	v2.0 with public API
Advanced HR (training LMS, recruitment ATS)	Core HR enough for v1.0; advanced HR feature creep	v1.2+ if demand
Marketing automation (email sequences, drip campaigns)	Marketing CRM focus on lead pipeline, not automation	v1.2 with marketing module expansion
Parent payment gateway EMI options	Requires NBFC partnerships	v1.2 with partner

12.2 Future Roadmap Placeholder

खालील items v1.0 च्या बाहेर आहेत पण product roadmap मध्ये tracked आहेत. हे items Vision v1.0 Section 13 (Long-Term Product Vision 5-10 Years) शी aligned आहेत:

- **v1.1 (Q1 2027):** Tally XML export, biometric integration improvements, custom report builder enhancements, transport management with partner, voice daily sheet pilot.
- **v1.2 (Q3 2027):** Multi-language expansion (Tamil, Telugu, Kannada), offline-first mobile, advanced HR (LMS), marketing automation, EMI options.
- **v2.0 (2028+):** Marketplace launch, parent community (opt-in), wearable integration, advanced AI (vision-based observations, predictive milestone alerts at scale).
- **v3.0 (2029+):** International expansion (SE Asia, Middle East, Africa), multi-currency, global compliance frameworks, third-party integrations marketplace.

13. Traceability Matrix (RTM)

Requirement Traceability Matrix (RTM) हा PreOne च्या requirements engineering चा final quality gate आहे. यात प्रत्येक requirement चा complete trace chain documented आहे — Vision → BR → FR → BRC → AC → Test Case → Implementation. हे traceability ensure करते की कोणताही requirement orphaned नाही आणि कोणताही implementation un-traced नाही.

RTM चा primary use case: (1) Impact analysis — BR change केला तर कोणते FRs, BRC rules, ACs affected होतील; (2) Coverage analysis — कोणत्या BRs ची ACs पूर्ण झाली आहेत; (3) Audit compliance — ISO 27001 / SOC 2 audits मध्ये traceability mandatory; (4) Onboarding — new team members ला context मिळतो. खालील master table 30 BRs चा trace chain दर्शवतो.

13.1 Requirements Flow

PreOne च्या requirements hierarchy खालीलप्रमाणे आहे. ही hierarchy top-down आहे — वरचा layer खालच्या layer ला drive करतो, आणि खालचा layer वरच्या layer चा verification provide करतो:

Vision → Business Requirement (BR) → Functional Requirement (FR) → Business Rule (BRC)
→ Acceptance Criteria (AC) → Test Case (TC) → Implementation

13.2 Master Traceability Matrix

BR ID	BR Title	FR IDs	BRC Category	AC IDs	Status
BR-001	Student Admission	FR-006 to FR-010	Eligibility & Approval	AC-005 to AC-008	Covered
BR-002	Parent Communication	FR-026 to FR-029	Communication	AC-022 to AC-025	Covered
BR-003	Daily Operations Digital	FR-020 to FR-025	Operational	AC-017 to AC-021	Covered
BR-004	Fee Management	FR-030 to FR-035	Financial	AC-026 to AC-030	Covered
BR-005	Inventory Management	FR-036 to FR-039	Inventory	AC-031 to AC-035	Covered
BR-006	Multi-School SaaS	FR-054 to FR-056	Platform	AC-050 to AC-053	Covered
BR-007	Lead Capture & CRM	FR-001 to FR-005	Eligibility & Notification	AC-001 to AC-004	Covered
BR-008	Student Lifecycle	FR-011 to FR-014	Eligibility	AC-009 to AC-012	Covered
BR-009	Academic Observations	FR-015 to FR-019	Academic	AC-013 to AC-016	Covered
BR-010	HR & Payroll	FR-040 to FR-043	HR	AC-036 to AC-039	Covered
BR-011	Compliance Management	FR-047	Compliance	AC-043	Covered
BR-012	Real-time Dashboards	FR-048 to FR-050	Platform & Notification	AC-044 to AC-046	Covered
BR-013	Notification Engine	FR-052	Notification	AC-048	Covered

BR ID	BR Title	FR IDs	BRC Category	AC IDs	Status
BR-014	Audit Trail	FR-056	Platform	AC-052	Covered
BR-015	Mobile Apps	FR-057, FR-058	Platform	AC-022, AC-044	Covered
BR-016	Multi-language Support	FR-059	Platform	AC-044	Covered (partial)
BR-017	Visitor Management	FR-045	Operational & Compliance	AC-041, AC-042	Covered
BR-018	Asset Management	FR-044, FR-046	Operational	AC-040	Covered
BR-019	Event Management	FR-029	Communication	AC-025	Covered
BR-020	Report Cards	FR-019	Academic	AC-016	Covered
BR-021	Refund Processing	FR-034	Financial & Approval	AC-030	Covered
BR-022	Bulk Import/Export	FR-060	Platform	AC-008	Covered
BR-023	Custom Workflows	FR-053	Platform	AC-049	Covered
BR-024	Biometric Attendance	FR-020 (integration)	Operational	AC-017	Covered (integration)
BR-025	AI Observation Assistant	FR-016 (AI feature)	Academic	AC-013	Covered
BR-026	Parent Feedback	FR-028 (survey)	Communication	AC-024	Covered
BR-027	Third-party Integrations	Future	Platform	Future	Deferred v1.1+

BR ID	BR Title	FR IDs	BRC Category	AC IDs	Status
BR-028	Transport Management	Future	Operational	Future	Deferred v1.1+
BR-029	Marketplace	Future	Platform	Future	Deferred v2.0+
BR-030	Voice Daily Sheet	Future	Operational	Future	Deferred v1.2+

13.3 Coverage Summary

RTM coverage analysis खालीलप्रमाणे आहे. हा summary identify करतो कोणते BRs v1.0 मध्ये fully covered आहेत आणि कोणते future versions कडे deferred आहेत:

Coverage Status	BR Count	% of Total	Notes
Fully Covered (v1.0)	26	87%	P0 + P1 BRs all covered with FRs + ACs
Partially Covered	1	3%	BR-016 Multi-language: 3 of 8 languages in v1.0
Deferred to v1.1+	2	7%	BR-027 Integrations, BR-028 Transport
Deferred to v2.0+	1	3%	BR-029 Marketplace
Deferred to v1.2+	1	3%	BR-030 Voice daily sheet
Total	30	100%	All 30 BRs traceable to FR/BRC/AC or deferred

13.4 Traceability Process

RTM चा maintenance खालील process द्वारे होतो. हा process software development lifecycle चा embedded step आहे — separate RTM update activity नाही:

- **Requirement Creation:** नवीन BR/FR/AC created करताना traceability links mandatory. बिना trace chain कोणताही requirement merged होणार नाही.
- **Sprint Planning:** Each sprint RTM review करतो — orphan requirements identified, missing ACs flagged.
- **Code Review:** PR template मध्ये "Related AC" field mandatory — code without AC reference rejected.
- **QA Cycle:** Test cases traceable to ACs; AC without test case flagged; coverage report auto-generated.
- **Release Sign-off:** Release readiness check: 100% P0 BRs covered, ≥95% P1 covered, 0 orphan ACs.
- **Quarterly Audit:** RTM completeness audit by Product team; gaps remediated within sprint.

Appendix A: Document Control

Document Control section या PRD च्या versioning, approval, distribution, आणि maintenance चा authoritative record आहे. हा section ISO 27001 / SOC 2 documentation requirements शी compliant आहे.

A.1 Version History

Version	Date	Author	Changes	Status
0.1	2026-06-01	Product Team	Initial draft — outline + sections 1-3	Draft
0.5	2026-06-15	Product Team	Added sections 4-7 (FR, NFR, BR, Roles)	Draft
0.8	2026-06-25	Product Team	Added section 8 (Module Requirements for 13 domains)	Draft

Version	Date	Author	Changes	Status
0.9	2026-07-01	Product Team	Added sections 9-13 + Appendix A; integrated BRC v1.0 references	Reviewed
1.0	2026-07-12	Product Team	Product Freeze — reviewed + approved by all stakeholders	Frozen

A.2 Approval Matrix

या PRD v1.0 च्या Product Freeze साठी खालील stakeholders ची approval mandatory होती. सर्व approvals documented आणि archived आहेत:

Role	Name	Department	Approval Date	Signature
Chief Product Officer	【CPO Name】	Product	2026-07-12	【Digital signature on file】
VP Engineering	【VP Eng Name】	Engineering	2026-07-12	【Digital signature on file】
Compliance Officer	【Compliance Name】	Legal & Compliance	2026-07-12	【Digital signature on file】
Business Head	【Business Head Name】	Business	2026-07-12	【Digital signature on file】
Operations Head	【Ops Head Name】	Operations	2026-07-12	【Digital signature on file】
Academic Advisor	【Academic Advisor Name】	Academic	2026-07-12	【Digital signature on file】

A.3 Review Cadence

PRD v1.0 freeze नंतर खालील review cadence follow केला जाईल. ही cadence ensure करते की document living artifact राहतो आणि stale होत नाही:

- **Quarterly Review:** Every quarter Product team reviews PRD for accuracy. Minor updates (clarifications, additional examples) push as v1.0.X.
- **Semi-Annual Major Review:** Every 6 months comprehensive review. New BRs/FRs added as v1.X.0. Stakeholder re-approval required.
- **Annual Major Release:** Once a year, major version bump (v2.0). Triggers re-baseline of BRC + companion docs.
- **Emergency Change:** Critical compliance / security changes can be pushed as hotfix with CPO + Compliance Officer approval.

A.4 Distribution List

या PRD ची access खालील teams ला granted आहे. Access control via PreOne internal docs portal; download + share restricted:

Team	Access Level	Purpose
Product Management	Edit + Comment	Owns PRD; updates + clarifications
Engineering (All)	Read + Comment	Implementation reference
QA	Read + Comment	Test case derivation
Design / UX	Read	UI/UX decisions alignment
Compliance & Legal	Read + Comment	Compliance review
Business / Sales	Read	Customer commitments alignment
Operations	Read	Operational readiness
Customer Success	Read	Customer onboarding reference
Executive Leadership	Read	Strategic alignment

A.5 Companion Documents

या PRD च्या effective use साठी खालील companion documents reference केले जातात. हे documents PreOne documentation portal वर available आहेत:

Document	Version	Relationship
PreOne Vision Document	1.0	Strategic context for this PRD
PreOne BPM (Business Process Map)	1.0	Process flows referenced in Section 8 (Module Requirements)
PreOne BRC (Business Rules Catalog)	1.0	176 rules referenced in Section 6 (Business Rules)
PreOne ADR (Architecture Decision Records)	1-12	Architecture decisions referenced throughout
PreOne Database Schema	v2	Data model for implementation
PreOne API Specification	v1 (OpenAPI)	API contracts referenced in Section 8
PreOne Design System	v1	UI/UX components for implementation
PreOne Test Strategy	v1	QA approach for ACs in Section 11